

K-Bound: When Is Label-Free Adaptation Knowable?

Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

Pratik Niroula
Minnesota State University, Mankato
`pratik.niroula@mnsu.edu`

Abstract

Test-time adaptation updates a deployed model without target labels, but an update can help or harm. K-Bound asks the decision that comes first: when does label-free evidence justify replacing a frozen model with an adapted candidate? It allows three actions—ADAPT, FREEZE, and ABSTAIN—for the benefit $\Delta = R_T(f_0) - R_T(f_a)$. In a binary disagreement model, $\Delta = 2\mu_T(D)(M + \gamma)$ separates an observable margin M from hidden calibration drift γ . If the declared class satisfies $|\gamma| \leq \beta$, a strict population action is uniformly supportable exactly when $|M| > \beta$. Throughout the closed band $|M| \leq \beta$, the same label-free evidence does not uniformly support either strict action. Without added labels or a stated coupling, the same label-free evidence cannot shrink the unresolved drift budget. The practical method, Knowability-Guided Adaptation (KGA), is separate: it predicts Δ and acts only when $\hat{\Delta} \pm \varepsilon$ excludes zero. Here ε is an empirical uncertainty radius, not an estimate of β . On controlled CIFAR-10-C, reconciled results give point gains for Tent and EATA. Tent also has positive ordinary intervals over six corruption families, but its preregistered six-comparison Holm values are 0.09375, so we make no confirmatory win claim. On the locked CCT-20 camera-location shift, 5 checkpoints across 9 locations produced 45 cells. KGA made 44 FREEZE, 1 ABSTAIN, and 0 ADAPT decisions. It matched always freeze, and its mean accuracy was 0.18190227 higher than always adapt. It did not support the predeclared stronger claim. This is a safe-utility result, not evidence of selective natural-shift routing. K-Bound is a validity and rollback layer, not a universal accuracy booster.

Keywords: test-time adaptation, safe adaptation, abstention, selective prediction, distribution shift, identifiability, label-free risk estimation

1 Introduction

1.1 From Test-Time Adaptation to Update Commitment

Test-time adaptation updates a model using an unlabeled deployment batch. Under a favorable shift, methods such as entropy minimization, filtering, regularization, and sharpness-aware updates can recover lost accuracy. Under an unfavorable shift, the same methods can amplify collapse, confirmation bias, or sensitivity to batch composition [1, 4, 5, 6]. The practical question is therefore not only how to adapt. Before changing the deployed model, a system must decide whether the proposed update should be accepted at all.

1.2 K-Bound in One View

Let f_0 be the frozen predictor and f_a a candidate produced in shadow state. We define adaptation benefit as

$$\Delta = R_T(f_0) - R_T(f_a). \tag{1}$$

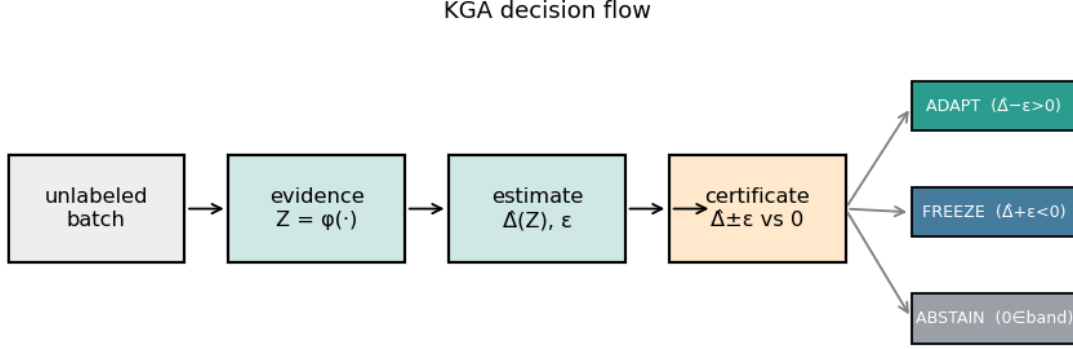


Figure 1: Practical KGA workflow. The adapter proposes f_a in shadow state; KGA evaluates label-free evidence, estimates benefit, and commits only if its finite-sample interval excludes zero. This interval rule is distinct from the population (M, γ, β) frontier.

Positive Δ means that adaptation helps; negative Δ means that it harms. K-Bound keeps uncertainty visible through three actions: ADAPT when the evidence supports benefit, FREEZE when it supports harm, and ABSTAIN when neither direction is clear. Freeze and abstain both retain f_0 , but they record different conclusions. Freeze supports a negative benefit. Abstention says only that the available evidence does not support a strict direction.

K-Bound has two layers. The first is a population result: given an explicit limit on hidden target drift, can any strict action be justified for every target distribution allowed by that limit? The second is a practical certificate: for the current deployment unit, is an estimated benefit far enough from zero to support an action? The first layer uses (M, γ, β) ; the second uses $(\hat{\Delta}, \epsilon)$. They study the same benefit but answer different questions. We keep them separate because an empirical uncertainty radius cannot establish a population drift bound.

1.3 Main Contributions

This compact paper makes five claims.

1. It formulates label-free TTA as an adapt/freeze/abstain decision about the paired risk difference in Eq. (1).
2. It gives an exact strict-commitment frontier over a declared calibration-drift class, with separate interior and boundary arguments, and an exact audit floor for what the same label-free evidence cannot identify.
3. It states a coverage-based finite-sample certificate that controls marginal false commitments, while distinguishing marginal from conditional error.
4. It presents KGA as an adapter-specific, rollback-capable wrapper and gives the direct logical bridge from the population functional to the empirical estimand.
5. It reports a source-hashed, reconciled empirical panel with controlled mixed regimes, natural shifts, and negative diagnostics under one decision rule and one metric vocabulary.

1.4 Empirical Scope and Nonclaims

A router can beat both fixed policies only when some updates help, others harm, and the evidence separates the two groups. Controlled CIFAR-10-C provides exact-rank point gains for Tent and EATA, while SAR is negative. In a retrospective current-policy sensitivity over six observed corruption families, Tent has positive ordinary family-bootstrap intervals against both fixed policies. The within-Tent two-contrast Holm values are post hoc, however, and the preregistered six-comparison Holm adjustment gives $p = 0.09375$ for both Tent contrasts. We therefore make no cluster-robust or confirmatory win claim. Office-Home, Camelyon17, and RxRx1 mainly provide descriptive no-harm or one-sided evidence. iWildCam remains withheld pending an official-metric, population-sealed rerun. PACS, ImageNet-R, CIFAR-10.1, and several ImageNet-C candidate rows are null or negative. These rows remain visible because K-Bound is a validity and rollback layer, not a universal accuracy booster.

Sections 3–6 define the decision and its two theoretical layers. Sections 7–8 describe the evidence and results. The remaining sections explain sensitivity, limitations, reproducibility, and the exact scope of each claim.

2 Related Work

2.1 Test-Time Adaptation

Tent, EATA, SAR, CoTTA, MEMO, SHOT, and test-time training improve the update mechanism through entropy minimization, filtering, regularization, temporal stabilization, augmentation, or auxiliary objectives [1, 4, 5, 6, 7, 2, 3]. These methods primarily address the quality of f_a . K-Bound treats f_a as a candidate and addresses whether the candidate should replace f_0 .

2.2 Selective Adaptation, Guarding, and Monitoring

POEM and sequential risk monitors protect or inspect adaptation streams using betting, proxy-risk, or alarm mechanisms [12, 8, 11]. These are complementary to a pre-commitment decision. KGA evaluates a shadow candidate, supports rollback, and exposes uncertainty as abstention before model state is changed.

Recent selective TTA work makes the overlap sharper. Informed Adaptation learns topological features that predict whether an update will help [9]. This work rules out a novelty claim based only on learning when to adapt. KGA is operationally a learned router, not the first selective TTA gate.

2.3 Label-Free Performance and Benefit Estimation

ATC, agreement-on-the-line, AETTA, and prediction-matrix statistics estimate target accuracy from unlabeled outputs [13, 16, 15, 17, 34, 33]. KGA uses related evidence but predicts the paired benefit $R_T(f_0) - R_T(f_a)$ rather than either risk in isolation. The difference matters near zero: an accurate point estimate can still be insufficient for a strict update decision.

2.4 Distribution-Shift Uncertainty and Robust Decisions

An MMD-credal approach derives target-risk intervals and an adapt/abstain criterion under RKHS and covariate-shift assumptions [10]. Its uncertainty set is built around target risk. K-Bound instead studies a paired benefit on evidence fibres: it asks whether the same label-free law can hide opposite

signs of $R_T(f_0) - R_T(f_a)$. The assumptions and guarantees are therefore complementary rather than interchangeable.

2.5 Abstention, Identifiability, and Risk Control

Selective classification and reject-option learning abstain on labels, while domain-adaptation theory and label-free risk estimation identify assumptions needed under shift [21, 52, 53, 19, 18, 20]. K-Bound abstains on an *update*. Its empirical interval layer uses the split-conformal and risk-control logic that a covered interval supports a calibrated one-sided decision [36, 37, 40, 41]. We do not claim new conformal machinery.

2.6 Positioning and Surviving Gap

Prior work can improve an adapter, select an update, estimate performance, or monitor a deployed stream. K-Bound’s narrower contribution is the validity structure around those operations. It separates population identifiability from finite-sample interval coverage, defines strict adapt/freeze/abstain semantics, and states what evidence is needed before a learned router can support a strict action.

3 Problem Formulation and Validity Scope

3.1 Deployment Setting

Let P_S denote the source law and P_T the deployment law. The frozen predictor is f_0 , the shadow candidate is f_a , and $R_T(f) = \mathbb{E}_{(X,Y) \sim P_T}[\ell(f(X), Y)]$ is target risk. Their paired difference is the benefit $\Delta = R_T(f_0) - R_T(f_a)$ from Eq. (1).

Source labels may be used during training and calibration. At deployment, the system sees the target batch $X_{1:m}$, predictions, logits, and update statistics, but not target labels. An adapter $\mathcal{A}$ creates a temporary candidate f_a from $(f_0, X_{1:m})$. Label-free evidence is any vector

$$Z = \phi(X_{1:m}, f_0, f_a)$$

that does not read target labels. Target labels may be revealed after the decision solely for offline auditing of regret and false commits.

3.2 Three-Way Decision Semantics

Definition 1 (Decision semantics). The action set is $\{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$. Adapt commits f_a . Freeze rejects f_a because the interval supports harm. Abstain rejects the update because the interval does not support a strict direction. Both non-adapt actions serve f_0 in the reported regret calculation.

For condition j , let R_{0j} and R_{aj} be the frozen and adapted risks. An offline oracle picks the smaller value. Policy regret is the extra risk relative to that oracle, averaged over the stated evaluation units. Always-freeze regret is therefore $\max(\Delta_j, 0)$, while always-adapt regret is $\max(-\Delta_j, 0)$. Lower regret is better.

Table 1: Validity obligations are separated by layer. No empirical quantity silently discharges a population assumption.

Layer	Object	Required support	Supports
Population	M, γ, β	declared target class; M fixed on the augmented-evidence fibre; class richness for necessity	uniform strict-commitment frontier
Estimator	$Z \mapsto \hat{\Delta}$	development support; fixed evidence schema; adapter-specific transfer	meaningful paired-benefit prediction
Certificate	$\hat{\Delta} \pm \varepsilon$	marginal interval coverage at the declared scoring unit	marginal false-adapt and false-freeze control
Evaluation	regret, FA_u , FA_c , actions	frozen protocol; declared evaluation unit; labels revealed only after decisions	empirical performance and diagnostic claims

3.3 False Commitments and Decision Exposure

The error metric tied to the certificate is the marginal false-adapt probability

$$\text{FA}_u = \mathbb{P}(g(Z) = \text{ADAPT}, \Delta \leq 0). \quad (2)$$

The conditional rate $\text{FA}_c = \mathbb{P}(\Delta \leq 0 \mid g(Z) = \text{ADAPT})$ is operationally useful but is not controlled by the theorem below. Decision coverage means $\mathbb{P}(g(Z) \in \{\text{ADAPT}, \text{FREEZE}\})$; it is not adapt rate.

These quantities answer different questions. For example, a controller that never adapts has zero false adaptations, but it has not shown that it can identify a helpful update. For that reason, we report the number of adapt decisions together with FA_u and regret.

3.4 Validity by Layer

3.4.1 Risk Alignment

Evidence is risk-aligned on a declared class when identical evidence cannot hide opposite benefit signs. This property comes from the class and the evidence definition, not from using a powerful regressor. Support checks and residual diagnostics may reveal a transfer failure, but they cannot prove uniform alignment.

3.5 Scope of the Population Theory

The population theorem below assumes binary labels and 0/1 loss. KGA itself only needs a scalar benefit target, so we also evaluate it on multiclass benchmarks. Those experiments do not extend the binary theorem. In multiclass problems, we may write $\Delta = \mathbb{P}_T(D)(p_a - p_0)$, where $p_a = \mathbb{P}_T(f_a = Y \mid D)$ and $p_0 = \mathbb{P}_T(f_0 = Y \mid D)$; no identity $p_0 = 1 - p_a$ is assumed outside binary classification.

3.6 Reading Guide

Six quantities carry the argument. M is the label-free margin seen on cases where the frozen and adapted models disagree. γ is the target calibration drift that labels would reveal, and β is the largest drift allowed by the declared population class. Their combined effect determines the true benefit Δ . In the practical method, $\hat{\Delta}$ estimates that benefit from label-free evidence, while ε is the uncertainty radius around the estimate. The population theory uses $(M, \gamma, \beta, \Delta)$; KGA uses $(\hat{\Delta}, \varepsilon)$ to decide whether the sign of Δ is clear enough to act. The distinctions that carry the paper are $M \neq \hat{\Delta}$ and $\beta \neq \varepsilon$.

4 Population Knowability

We now state the main population result. We repeat the setup so this short paper can be read on its own.

4.1 Disagreement-Region Model

Assumption 1 (Deployment setup). Let $Y \in \{0, 1\}$, use 0/1 loss, and assume $\mu_T(D) > 0$ for $D = \{x : f_0(x) \neq f_a(x)\}$. Let $s : \mathcal{X} \rightarrow [0, 1]$ be a source-calibrated candidate-correctness score and define the target pointwise correctness $\eta_a(x) = \mathbb{P}_T(f_a(X) = Y \mid X = x)$. On D , set

$$M = \mathbb{E}[s(X) \mid D] - \frac{1}{2}, \quad \gamma = \mathbb{E}[\eta_a(X) - s(X) \mid D]. \quad (3)$$

The declared population class is $\mathcal{C}_\beta = \{P_T : |\gamma(P_T)| \leq \beta\}$ for an externally supplied $\beta \geq 0$.

Remark 1 (Residual identity). The definition gives

$$\begin{aligned} M + \gamma &= (\mathbb{E}[s(X) \mid D] - \frac{1}{2}) + \mathbb{E}[\eta_a(X) - s(X) \mid D] \\ &= \mathbb{E}[\eta_a(X) \mid D] - \frac{1}{2}. \end{aligned}$$

This is an identity, not empirical evidence that calibration transfers.

Definition 2 (Risk alignment). Evidence Z is risk-aligned on a class $\mathcal{P}$ if no two laws in $\mathcal{P}$ with opposite benefit signs induce the same law of Z . Population frontier claims take M to be fixed on each augmented evidence fibre, equivalently by using $\tilde{Z} = (Z, M)$.

Definition 3 (Strict directional soundness). At a fixed augmented-evidence law, ADAPT is uniformly directionally sound when $\Delta(P) > 0$ for every compatible $P \in \mathcal{C}_\beta$; FREEZE is sound when $\Delta(P) < 0$ for every such law. A pointwise maximal three-way rule commits whenever one strict action is sound and abstains otherwise. The strict convention treats $\Delta = 0$ as insufficient for either directional claim, although the actions then have equal task risk.

4.2 Label-Kernel Freedom on the Disagreement Region

Lemma 1 (Label-kernel freedom on disagreement). *Fix $(\mu_T, P_S, f_0, f_a, s)$, hence D and M , and fix the conditional label law outside D . For every measurable $\eta : D \rightarrow [0, 1]$ there is a target law with input marginal μ_T and $\mathbb{P}(f_a(X) = Y \mid X = x) = \eta(x)$ on D . Every label-free observable formed from the input batch, the fixed models, and their outputs has the same law for all such η , while*

$$\gamma = \mathbb{E}[\eta(X) \mid D] - \frac{1}{2} - M.$$

Proof. On D , binary predictors disagree, so their outputs are 0 and 1. Assign conditional mass $\eta(x)$ to $Y = f_a(x)$ and $1 - \eta(x)$ to $Y = f_0(x)$, retaining the fixed kernel outside D . This changes no input marginal or model output and therefore changes no label-free evidence law. The expression for γ follows from Eq. (3). $\square$

4.3 Benefit Reduction and Matched-Evidence Impossibility

Lemma 2 (Disagreement-region reduction). *Under Assumption 1, with $\bar{a} := \mathbb{P}_T(f_a=Y \mid D)$,*

$$\begin{aligned} \Delta &= 2\mu_T(D)(\bar{a} - \frac{1}{2}), \\ \text{sign } \Delta &= \text{sign}(\bar{a} - \frac{1}{2}) = \text{sign}(M + \gamma). \end{aligned}$$

The identity $\text{sign } \Delta = \text{sign}(M + \gamma)$ holds for every target law; it does not assume $|\gamma| \leq \beta$.

Proof. On D exactly one of f_0, f_a is correct under 0/1 loss, so the benefit equals $2\mu_T(D)(\bar{a} - \frac{1}{2})$. By the definitions of M and γ ,

$$\begin{aligned} M + \gamma &= \left(\mathbb{E}_{\mu_T|D}[s] - \frac{1}{2} \right) + \mathbb{E}_{\mu_T|D}[\eta_a - s] \\ &= \mathbb{E}_{\mu_T|D}[\eta_a] - \frac{1}{2} = \bar{a} - \frac{1}{2}, \end{aligned}$$

so $\text{sign } \Delta = \text{sign}(\bar{a} - \frac{1}{2}) = \text{sign}(M + \gamma)$. $\square$

Theorem 1 (Interior matched-evidence impossibility). *Fix $\beta > 0$. For any margin with $|M| < \beta$ there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ such that $\text{Law}(\tilde{Z} \mid P_T^1) = \text{Law}(\tilde{Z} \mid P_T^2)$ for $\tilde{Z} = (Z, M)$, $M(P_T^1) = M(P_T^2) = M$, and $\text{sign } \Delta_1 = -\text{sign } \Delta_2 \neq 0$. (For $\beta = 0$ the class $\mathcal{C}_0$ forces $\gamma = 0$, so $\text{sign } \Delta = \text{sign } M$ is determined and no such pair exists: matched-evidence ambiguity is a strictly-positive-drift phenomenon.)*

Proof. This is Lemma 1 read at one radius. Choose $0 < \delta < \min\{\beta - |M|, \frac{1}{2}\}$ and apply the lemma with the two constant kernels $\eta \equiv \frac{1}{2} + \theta\delta$, $\theta \in \{+1, -1\}$, holding the off- D kernel K° common to both worlds. The lemma supplies the two laws P_T^θ , and: *class membership*, $\bar{a}_\theta = \frac{1}{2} + \theta\delta$ so $\gamma_\theta = \theta\delta - M$ with $|\gamma_\theta| \leq \delta + |M| < \beta$, hence both worlds lie in $\mathcal{C}_\beta$; *matched evidence*, $\text{Law}(W \mid P_T^\theta)$ does not depend on η by the lemma and $\tilde{Z} = (Z, M)$ is a function of W , so $\text{Law}(\tilde{Z} \mid P_T^+) = \text{Law}(\tilde{Z} \mid P_T^-)$; *opposite signs*, Lemma 2 gives $\Delta_\theta = 2\mu_T(D)\theta\delta \neq 0$ with $\text{sign } \Delta_\theta = \theta$. $\square$

The two evidence laws here coincide *exactly*, not merely approximately: the construction changes only the label kernel on D , and Z is a function of unlabeled inputs and model outputs. We therefore do not present this as a Le Cam two-point argument. A Le Cam argument trades off a positive total-variation distance against the achievable error; here the distance is zero, which is the degenerate and strongest case, and the corresponding statement is exact non-identifiability rather than a rate.

Corollary 1 (Matched-evidence abstention lower bound). *Fix $\alpha \in (0, \frac{1}{2})$. On the shared evidence law constructed in Theorem 1, any possibly randomized label-free rule g satisfying $\mathbb{P}[g = \text{ADAPT}, \Delta \leq 0] \leq \alpha$ in every admissible world and $\mathbb{P}[g = \text{FREEZE}, \Delta \geq 0] \leq \alpha$ in every admissible world must satisfy $\mathbb{P}[g = \text{ABSTAIN}] \geq 1 - 2\alpha$.*

Proof. The rule has the same action probabilities in the two evidence-identical worlds. Its adapt probability is at most α in the negative world and its freeze probability is at most α in the positive world. The three action probabilities sum to one. $\square$

4.4 Boundary Case

Proposition 1 (Boundary case and closed-band abstention). *Fix $\alpha \in (0, \frac{1}{2})$. Under Definition 3, abstention is the pointwise maximal sound three-way action throughout the closed band $|M| \leq \beta$, and any label-free rule controlling both marginal errors at level α abstains there with probability at least $1 - 2\alpha$. Opposite nonzero signs are asserted only in the interior $|M| < \beta$.*

Proof. For $|M| < \beta$ this is Theorem 1 together with Corollary 1. For $|M| = \beta > 0$, the declared class contains an augmented-evidence-identical zero-benefit world (take $\gamma = -M$, admissible since $|\gamma| = \beta$) and a strict-benefit world of sign M (take $\gamma = 0$). In the zero-benefit world $\Delta = 0$, so *both* directional error events $\{\text{ADAPT}, \Delta \leq 0\}$ and $\{\text{FREEZE}, \Delta \geq 0\}$ occur whenever the rule commits; since the rule's action probabilities are identical across the two evidence-identical worlds, dual control at level α gives $\mathbb{P}[\text{ADAPT}] \leq \alpha$ and $\mathbb{P}[\text{FREEZE}] \leq \alpha$, hence $\mathbb{P}[\text{ABSTAIN}] \geq 1 - 2\alpha$, and neither strict action is uniformly directionally sound. For $M = \beta = 0$ the class $\mathcal{C}_0$ forces $\gamma = 0$, so $\Delta \equiv 0$ and the same conclusion holds directly. $\square$

This directional convention is stricter than task regret: at $\Delta = 0$, adapt and freeze are risk-equivalent but neither certifies a strict sign.

4.5 Exact Strict-Commitment Frontier

Theorem 2 (Exact strict-commitment frontier). *Under Assumption 1, fix the observables $(\mu_T, P_S, f_0, f_a, s)$ so M is determined, and let $\beta \geq 0$. The necessity and maximality clauses use the full class $\mathcal{C}_\beta$ and therefore its richness: it contains the evidence-identical label kernels constructed in Theorem 1 and the zero-benefit boundary law. For a narrower declared subclass not closed under these constructions, only the sufficiency clause follows without an additional richness assumption.*

- (i) **Reduction.** *For every P_T , $\text{sign } \Delta = \text{sign}(M + \gamma)$ (Lemma 2).*
- (ii) **Sufficiency.** *If $|M| > \beta$ and $P_T \in \mathcal{C}_\beta$, then $\text{sign } \Delta = \text{sign } M$.*
- (iii) **Necessity.** *If $|M| < \beta$ (which requires $\beta > 0$), there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ with the same augmented-evidence law and opposite nonzero benefit signs; at $|M| = \beta > 0$ the admissible drift $\gamma = -\text{sign}(M)\beta$ yields $\Delta = 0$, and in the degenerate case $M = \beta = 0$ the class $\mathcal{C}_0$ forces $\Delta \equiv 0$. Hence no strict benefit direction is uniformly supported by $\text{Law}(\tilde{Z})$ over $\mathcal{C}_\beta$ whenever $|M| \leq \beta$.*
- (iv) **Strict-commitment iff.** *At the fixed augmented-evidence law, a strict action in $\{\text{ADAPT}, \text{FREEZE}\}$ is uniformly directionally sound in the sense of Definition 3 if and only if $|M| > \beta$; on the closed band $|M| \leq \beta$ (including $M = \beta = 0$, where $\Delta \equiv 0$) abstention is the pointwise maximal sound action.*

The population rule $M > \beta \Rightarrow \text{ADAPT}$, $M < -\beta \Rightarrow \text{FREEZE}$, $|M| \leq \beta \Rightarrow \text{ABSTAIN}$ is the pointwise maximal sound certificate on $\mathcal{C}_\beta$.

Proof. Part (i) is Lemma 2. For (ii), if $M > \beta$ and $|\gamma| \leq \beta$, then $M + \gamma > 0$, so $\text{sign } \Delta = \text{sign}(M + \gamma) = +1$; the negative case is symmetric. For (iii), Theorem 1 supplies two laws in the same augmented-evidence fibre with opposite nonzero signs whenever $|M| < \beta$. At $|M| = \beta$ the boundary law with $\gamma = -\text{sign}(M)\beta$ gives $\Delta = 0$, still precluding a determined nonzero sign. Part (iv) combines (ii), (iii), and Proposition 1. Pointwise maximality follows because (ii) certifies the unique strict direction outside the band, while Theorem 1 and Proposition 1 rule out both strict directions inside and on the boundary. $\square$

Region	Meaning	Action
$M > \beta$	drift cannot flip a positive sign	ADAPT
$M < -\beta$	drift cannot flip a negative sign	FREEZE
$ M < \beta$	drift can reverse the strict sign	ABSTAIN
$ M = \beta > 0$	drift can reduce benefit to zero	ABSTAIN
$M = \beta = 0$	benefit is identically zero	ABSTAIN

4.6 Interpretation and Limits of the Frontier

The theorem depends on a declared class. Choosing $\beta = 0$ assumes that calibration does not drift; it is not a conservative default. Without a credible upper bound on $|\gamma|$, the population frontier cannot support a robust strict action. More unlabeled samples do not solve this problem because the missing difference lies in the target labels, not in the input batch.

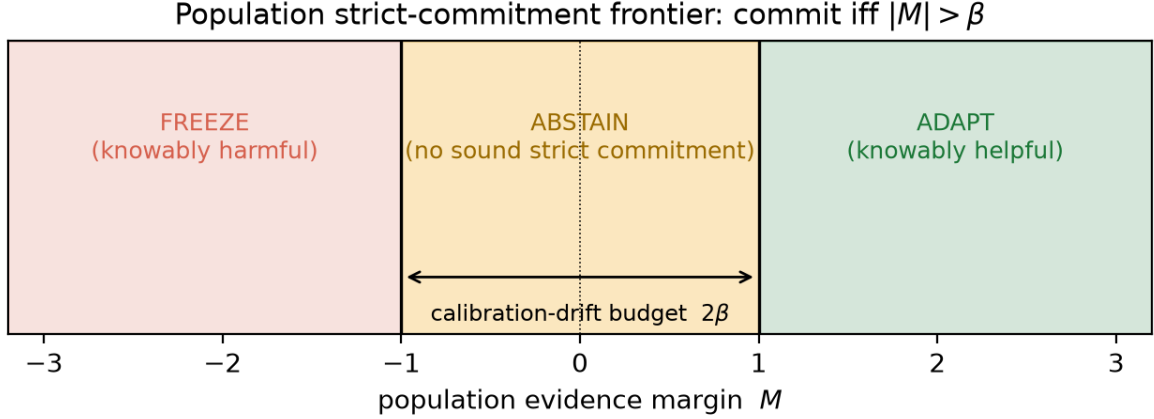


Figure 2: Population strict-commitment frontier. $M > \beta$ supports adapt; $M < -\beta$ supports freeze; and $|M| \leq \beta$ supports no uniform strict commitment. Interior ambiguity uses opposite nonzero benefit signs, whereas the boundary uses zero-benefit ambiguity. This population rule is not the numerical KGA rule used on the real-data panels.

Empirical abstention is not automatically evidence of structural unknowability. A wide finite-sample interval may instead reflect limited calibration data, estimator error, or transfer failure. The frontier therefore leaves one central question: can its required drift budget be learned from the same label-free evidence?

5 Can the Drift Budget Be Learned Label-Free?

5.1 Audit Setup and Evidence Fibres

Let W collect everything a deployment-time audit may read: the unlabeled target batch, the fixed models and score function, label-free statistics derived from them, and any source or calibration sample whose law is specified independently of the target label kernel. An audit is a nonnegative random quantity $\hat{\beta} = g(W, U)$, where U is independent randomization.

For a declared class $\mathcal{C}$ and an observable law $z = \text{Law}(W)$, define the evidence fibre

$$\mathcal{C}(z) = \{P \in \mathcal{C} : \text{Law}(W \mid P) = z\}$$

and its unresolved drift radius

$$\Gamma_z(\mathcal{C}) = \sup_{P \in \mathcal{C}(z)} |\gamma(P)|. \quad (4)$$

The fibre collects target laws that the audit cannot distinguish. Its radius is therefore the smallest uniform drift budget that those observations could support.

5.2 Exact Minimax Label-Free Budget

Theorem 3 (Exact label-free audit floor). *Fix a nonempty fibre $\mathcal{C}(z)$ and $\delta \in [0, 1)$. If $\hat{\beta}$ satisfies $\mathbb{P}_P(|\gamma(P)| \leq \hat{\beta}) \geq 1 - \delta$ for every $P \in \mathcal{C}(z)$, then*

$$\mathbb{P}(\hat{\beta} \geq \Gamma_z(\mathcal{C})) \geq 1 - \delta.$$

The constant audit $\hat{\beta} \equiv \Gamma_z(\mathcal{C})$ is valid with $\delta = 0$. Consequently, on the high-probability event above, an audited frontier commits only where the frontier using the declared radius $\Gamma_z(\mathcal{C})$ already commits.

Proof. Every law in the fibre induces the same law for W , and hence the same law for $\hat{\beta}$. Validity for each fixed P requires that common law to exceed $|\gamma(P)|$ with probability at least $1 - \delta$. Taking an increasing sequence of attainable drifts approaching the supremum gives the lower bound. The constant audit is valid by the definition of the supremum. If $\hat{\beta} \geq \Gamma_z$, then $|M| > \hat{\beta}$ implies $|M| > \Gamma_z$. $\square$

This result does not say that useful budgets never exist. It says that the same label-free fibre cannot create information about the target label kernel that the fibre definition removes. A valid audit must cover the unresolved radius; reading more functions of the same W does not shrink it.

5.3 Declared Budget and Abstention Radius

For the full declared class $\mathcal{C}_\beta = \{P : |\gamma(P)| \leq \beta\}$ with $0 \leq \beta \leq \frac{1}{2}$, the label-kernel construction in Lemma 1 gives

$$\Gamma_z(\mathcal{C}_\beta) = \beta. \quad (5)$$

Thus the least fibre-wide budget a label-free audit can certify is the budget already declared by the class. The same number is also the population abstention radius:

least valid fibre-wide drift budget = population abstention radius = β

This equality connects the audit problem to the strict-commitment frontier. It does not turn the declaration into an estimate.

5.4 Why Labels at the Wrong Domain Buy Nothing

Suppose the audit also observes a fully labeled calibration domain, but the declared class contains no coupling between that domain's label kernel and the deployment label kernel. The construction in Lemma 1 changes only deployment labels on D and leaves the calibration sample unchanged. The augmented evidence law is therefore identical across the same target fibre, so Theorem 3 applies without a smaller radius. Labels help only when they come from the deployment domain or when a stated assumption links the labeled and deployment domains.

5.5 What Additional Structure Can Buy

The missing information can be supplied, but each route changes the problem. Deployment labels can estimate target correctness directly. Historical labeled deployments can support an episode-marginal budget under an exchangeability model. A structural coupling can bound the change between calibration and target domains. Domain knowledge can declare a physically meaningful limit. These are purchases made with labels or assumptions; they are not consequences of the current unlabeled batch. Detailed rate results and extended constructions belong to the long-form theory record and are not required by the empirical claims here.

5.6 Implications for K-Bound

KGA does not estimate β and does not numerically apply $|M| > \beta$ on the reported real-data benchmarks. Its practical certificate instead places a finite-sample uncertainty radius ε around the learned benefit $\hat{\Delta}$. The population frontier and KGA therefore share the target Δ , but they answer different questions with different uncertainty objects.

6 Finite-Sample Certification and KGA

6.1 Finite-Sample Adapt/Freeze/Abstain Certificate

The practical layer estimates the realized benefit directly. We import the shared certificate statement and proof.

Theorem 4 (Finite-sample adapt/freeze/abstain certificate). *Suppose Δ , $\hat{\Delta}$, and the nonnegative random radius ε are measurable on a common probability space containing all calibration, benefit-model fitting, and deployment-unit randomness, and that the benefit interval satisfies marginal coverage*

$$\mathbb{P}[|\hat{\Delta} - \Delta| \leq \varepsilon] \geq 1 - \alpha.$$

Define

$$g(Z) = \begin{cases} \text{ADAPT} & \hat{\Delta} - \varepsilon > 0, \\ \text{FREEZE} & \hat{\Delta} + \varepsilon < 0, \\ \text{ABSTAIN} & \text{otherwise.} \end{cases}$$

Then, marginally over that common probability space,

$$\begin{aligned} \mathbb{P}[g(Z) = \text{ADAPT}, \Delta \leq 0] &\leq \alpha, \\ \mathbb{P}[g(Z) = \text{FREEZE}, \Delta \geq 0] &\leq \alpha. \end{aligned}$$

Proof. On the event $\{|\hat{\Delta} - \Delta| \leq \varepsilon\}$, the condition $\hat{\Delta} - \varepsilon > 0$ implies $\Delta > 0$, so a false adapt requires the complement of this event, which has probability at most α . The freeze case is symmetric. $\square$

Exchangeability, or an explicit shift correction, is one route to this coverage assumption. Risk alignment is not required for the conditional-on-coverage implication.

Remark 2 (Marginal vs. conditional error). Theorem 4 bounds the *marginal, unconditional* false-adapt and false-freeze probabilities (FA_u). It does *not* bound the conditional rate $\text{FA}_c = \mathbb{P}[\Delta \leq 0 \mid \text{ADAPT}]$, which we report empirically. Risk alignment (Definition 2) is required for the *frontier* identifiability claim; coverage alone certifies the *interval* rule above.

Remark 3 (Fallback when assumptions are unsupported). If interval coverage—or the calibration assumptions supporting it (exchangeability, or an explicit shift correction)—cannot be justified on the deployment batch, Theorem 4 supplies no level- α protection for either strict action. A deployment must then use a separately declared operational fallback, such as abstention or serving the frozen model; this is a policy recommendation, not a conclusion that either fallback is optimal. If instead *risk alignment* (Definition 2) cannot be justified, the population-frontier interpretation and the transferability of the benefit estimator are unsupported, even though the interval implication of Theorem 4 remains valid *conditional on coverage* (cf. Theorem 1).

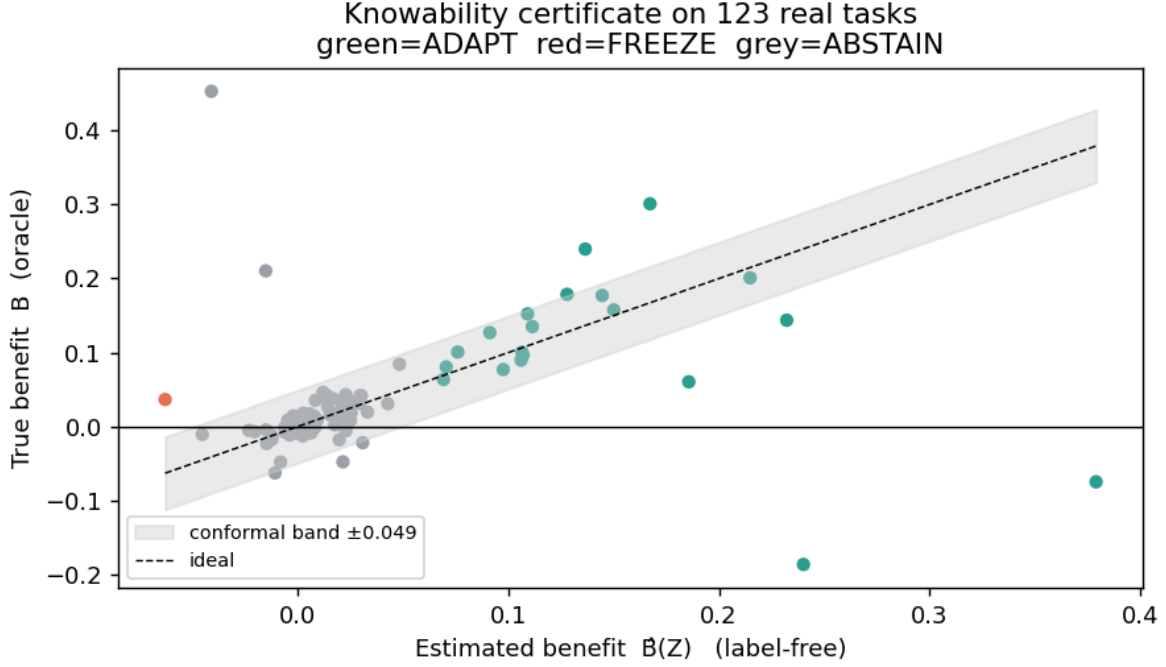


Figure 3: Empirical certificate. KGA adapts only when $\hat{\Delta} - \varepsilon > 0$, freezes only when $\hat{\Delta} + \varepsilon < 0$, and abstains otherwise. This finite-sample rule is distinct from the population rule in Fig. 2.

6.2 Population-to-Algorithm Bridge

Under the binary setup, the common target functional links the layers:

$$\underbrace{\Delta}_{\text{common target}} = 2\mu_T(D)(M + \gamma), \quad (6)$$

$$\underbrace{\hat{\Delta}}_{\text{KGA output}} = h_\theta(Z).$$

The population class bounds the first expression through $|\gamma| \leq \beta$. KGA bounds the same Δ through prediction error. If

$$\mathbb{P}(|\hat{\Delta} - \Delta| \leq \varepsilon) \geq 1 - \alpha, \quad (7)$$

then an interval that excludes zero supports the matching action for the observed deployment unit. The two guarantees have different scopes. The frontier is uniform over $\mathcal{C}_\beta$ at fixed population evidence. The interval statement is marginal over the declared calibration and deployment randomness.

Three limits matter. First, KGA does not estimate M , γ , or β in the reported real-data experiments. Second, ε covers total prediction error; it does not estimate or separate latent drift. Third, abstention can reflect structural ambiguity, limited data, a weak estimator, transfer failure, or conservative calibration. Abstention alone does not prove that the population law lies in $|M| \leq \beta$.

The release keeps this separation in code. The population API requires an externally declared (M, β) and fails closed when β is absent. The KGA API instead accepts $(\hat{\Delta}, \varepsilon)$ and has no β argument. A controlled algebraic check records the two actions side by side on cells with the same Δ . The actions may agree or disagree because the two layers use different uncertainty objects and scopes. This check is not a real-data estimate of β or a benchmark win.

Table 2: The direct bridge shares Δ but does not identify unlike quantities.

Quantity	Layer	Role
M	population	observable disagreement margin
γ	population	latent calibration drift
β	population	declared bound on $ \gamma $
Δ	both	target risk reduction
$\hat{\Delta}$	empirical	learned estimate from Z
ε	empirical	residual-coverage radius

Algorithm 1 Knowability-Guided Adaptation (KGA)

Require: Frozen model f_0 , adapter $\mathcal{A}$, unlabeled batch X , fitted h_θ , radius ε

- 1: Checkpoint f_0 and propose $f_a \leftarrow \mathcal{A}(f_0, X)$ in shadow state
 - 2: Evaluate f_0, f_a on X and extract $Z = \phi(X, f_0, f_a)$
 - 3: $\hat{\Delta} \leftarrow h_\theta(Z)$; $L \leftarrow \hat{\Delta} - \varepsilon$; $U \leftarrow \hat{\Delta} + \varepsilon$
 - 4: **if** $L > 0$ **then**
 - 5: commit f_a and return ADAPT
 - 6: **else if** $U < 0$ **then**
 - 7: roll back and return FREEZE
 - 8: **else**
 - 9: retain f_0 , log uncertainty, and return ABSTAIN
 - 10: **end if**
-

6.3 KGA Algorithm and Shadow-State Workflow

The workflow isolates candidate generation from commitment. No update reaches the serving state until the evidence schema, calibrated estimator, interval, and action have all been evaluated. A freeze or abstain decision discards the shadow state and retains the checkpointed model.

6.4 Evidence and Benefit Model

The controlled-grid implementation records 11 label-free coordinates with effective rank 9. Two coordinates are deterministic differences: `entropy_drop=pre_entropy-post_entropy` and `pbal_drop=pre_pbal-post_pbal`. The panel therefore contains 11 stored values, not 11 independent information sources. Together they describe confidence, class balance, change from the frozen model, and update size.

Natural-shift artifacts do not all use the same schema. iWildCam, RxRx1, and most archived tracks use the 11-feature panel. Camelyon17 uses 17 features by adding disagreement, entropy gap, free-energy shift, batch-statistic divergence, a source-fixed ATC estimate, and confidence drop. Office-Home uses 18 features and also records source-prior KL and mean candidate disagreement. A model and its radius are valid only for the schema on which they were calibrated. Missing, reordered, or invalid features force non-commitment.

The benefit model is a scikit-learn `GradientBoostingRegressor` with squared-error loss, 250 trees, depth 2, learning rate 0.05, subsample 0.8, and random seed 0. We refit it for each dataset and adapter. Thus “adapter agnostic” describes the interface: any adapter can propose a candidate that is evaluated and either committed or rolled back. It does not mean that a benefit model fitted for Tent transfers validly to EATA, SAR, or another adapter.

6.5 Calibration Designs

For a clean calibration set with residuals $r_i = |\hat{\Delta}_i - \Delta_i|$, sort $r_{(1)} \leq \dots \leq r_{(n)}$ and use

$$k = \lceil (n+1)(1-\alpha) \rceil, \quad \varepsilon = \begin{cases} r_{(k)}, & k \leq n, \\ +\infty, & k > n. \end{cases} \quad (8)$$

Under exchangeability, this is the ordinary split-conformal route to Eq. (7). The controlled grids use leave-one-condition-out, out-of-fold benefit predictions and leave-one-cell-out empirical residual radii. That prevents direct in-fold leakage, but it is not relabeled as exact split conformal or jackknife+. Its coverage is empirical and depends on stability. Natural-shift protocols instead separate development fitting, residual calibration, and target scoring where the saved protocol supports that separation. Each canonical row retains its own scope label because a numerical replay is not automatically a transfer proof.

The intuition is simple. Calibration records how far past benefit predictions missed their true values. The radius ε is a high empirical quantile of those misses. KGA acts only when the new prediction is farther from zero than this radius. A wider radius produces fewer decisions; a narrower radius produces more decisions but needs stronger evidence that calibration transfers.

Concrete example. Suppose KGA predicts a benefit of 0.04 with radius 0.01. The interval is $[0.03, 0.05]$, so every covered value is positive and KGA adapts. If the prediction is -0.04 with the same radius, the interval is $[-0.05, -0.03]$ and KGA freezes. If the prediction is 0.005, the interval is $[-0.005, 0.015]$; because it crosses zero, KGA abstains. The rule depends on the whole interval, not only on the point estimate.

6.6 Deployment Checks and Cost

The deployed workflow checkpoints active state, proposes adaptation on a temporary copy, evaluates both predictors, extracts evidence, computes the interval, commits or rolls back, and logs the action. It must fail closed when the evidence schema, checkpoint, adapter configuration, calibration identity, or numerical checks differ from the calibrated contract. In a continual stream, the last accepted model may become the new baseline only if the deployment policy states this rule in advance.

KGA is not “one extra forward pass.” Its total cost includes candidate adaptation (including backward computation), frozen inference, candidate inference, evidence extraction, benefit-model evaluation, a temporary model copy, and rollback. The tree and interval decision are lightweight, but creating the shadow candidate is not. We report no common latency number because the canonical manifest does not contain a comparable measurement across all tracks.

7 Experimental Design

7.1 Controlled and Natural Evidence

The evaluation has two branches. Controlled grids vary corruption, severity, batch composition, and update strength. This creates cells in which adaptation helps and cells in which it harms. The grids test whether KGA can separate those cases, although they use cross-fitted cells rather than a completely untouched target dataset. Natural and semantic shifts test transfer from development domains to target conditions. Many of these shifts are one-sided, so they may test no-harm behavior without giving any router a realistic chance to beat both fixed policies.

Table 3: Evaluation roles. A dataset name is not a claim: the candidate, calibration unit, replay status, and shift geometry determine what the row can support.

Track	Shift geometry	Candidates in canonical panel	Defensible use
CIFAR-10-C	controlled mixed corruption/composition	Tent, EATA, SAR	routing and candidate dependence; empirical LOO calibration
ImageNet-C	large-scale corruption grid	Tent, EATA, SAR	candidate-dependent scale diagnostic
Office-Home	semantic domain shift	locked candidate	primary no-harm, 54-cell stream-seed replication, and an invalidated exploratory five-checkpoint route
iWildCam	geographic/camera shift	locked candidate	numerical row withheld; official WILDS-metric rerun required
Camelyon17	hospital shift	OOD Tent/EATA/SAR agnostic	row; di- archived opened one-sided OOD diagnostic plus within-seed diagnostics
RxRx1	laboratory/batch shift	locked candidate	displayed model-seed-0 no-harm row plus a three-checkpoint all-freeze replication
PACS	domain generalization	aggregate candidate	negative aggregate diagnostic; incomplete decision replay
ImageNet-R	rendition and architecture panel	ten-backbone panel	negative architecture diagnostic, not one deployed policy
CIFAR-10.1	natural resampling	locked replay	negative low-margin cross-seed diagnostic

7.2 Evaluation Roles and Regime Map

Table 3 separates benchmark names from the claims that each archived protocol can support.

7.3 Metrics, Units, and Leakage Control

We report regret to the per-condition oracle, marginal and conditional false-adapt rates, adapt rate, the full action count, and decision coverage. Always-adapt and always-freeze have coverage one because they never abstain. If KGA has lower regret than both, the routing decisions were useful on that panel. This ordering alone does not prove interval coverage or population risk alignment.

Each track states its fitting unit, residual pool, and scoring unit. For a controlled cell, the estimator excludes that cell from fitting, and the radius excludes its residual. Natural rows use the archived development and target partitions. The gate never sees labels used to compute Δ ; those labels are revealed only after the action is fixed. We also keep model seeds, test-stream seeds, and repeated cells separate because they represent different kinds of uncertainty.

7.4 Canonicalization and Provenance

The generated table below is the canonical point-estimate table used by this compact submission. It is a post-hoc reconciled audit panel built from 106 compact source artifacts whose original and released SHA-256 hashes are recorded in `experiments/kbound/results/reconciled_panels_v1/source_manifest.json`. The corresponding JSON records the decision rule, $\alpha = 0.10$, generator identity, package versions, and panel-specific scope. Candidate rows remain separate. Historical interpolated-quantile results, superseded ImageNet-C grids, and development win-search artifacts are not merged into this table.

7.5 Inference Units, Clustering, and Multiplicity

The resampling unit must match the claim. A condition-level interval treats controlled cells as the units. A corruption-family interval resamples groups of related corruptions. Seed, backbone, domain, and physical session units answer different questions and are not interchangeable. Throughout this paper, a paired regret gap is defined as baseline regret minus KGA regret, so a positive value favors KGA. Confidence intervals follow the same sign convention.

The repository also contains exploratory searches conducted before the present reconciliation. Their historical counts were not frozen with one reproducible family definition, and two archived censuses disagree. We therefore do not promote a multiplicity total or call the full panel pre-registered. The release is a frozen claim-eligible panel assembled post hoc; prospectively locked, retrospective, historical, invalidated, and withheld rows are identified separately.

8 Results

The results answer three questions. First, can KGA beat both fixed policies when helpful and harmful updates coexist? Second, does that result hold across adapters and resampling units? Third, does the same behavior transfer to natural shifts? The answers are: yes at the exact-rank point-estimate level for controlled Tent and EATA; positive ordinary six-family intervals for Tent, but no candidate clears the preregistered six-comparison Holm gate; and mostly no-harm, negative, or withheld evidence on natural shifts.

8.1 Controlled Mixed Regime: CIFAR-10-C

8.1.1 Candidate-Specific Results: Tent, EATA, and SAR

The controlled stress grid is the clearest demonstration of routing value. For Tent, KGA regret is 0.0016, compared with 0.0080 for always-adapt and 0.1239 for always-freeze. For EATA the corresponding values are 0.0013, 0.0033, and 0.1313. Both beat the fixed policies by point estimate and have zero observed FA_u in the canonical table. Reconciliation from manifest 391983f4fa09d65c256ab8c620fea772866b652c219dad0c36574ce38c740482 gives Tent actions 1,107/359/694 and EATA actions 1,241/132/787 for adapt/freeze/abstain. The older 1,113/1,244 adapt counts are not canonical. The current-policy retrospective analysis averages run seeds and condition cells within each of six corruption families. With baseline regret minus KGA regret, Tent has gaps 0.00633 [0.00254, 0.00954] against always-adapt and 0.12229 [0.06331, 0.17772] against always-freeze; these are ordinary, unadjusted 95% family-bootstrap intervals. The exact one-sided sign-flip p -values are 0.03125 and 0.015625. Adjusting only these two Tent contrasts gives 0.03125 for each, but that within-candidate family is post hoc. The preregistered family contains all six candidate-by-baseline contrasts; its Holm-adjusted Tent values are 0.09375 for both comparisons and fail at 0.05. EATA's adapt-side family interval $[-0.00052, 0.00438]$ also crosses zero, and SAR loses to always-adapt. We therefore retain Tent and EATA as exact-rank point results and describe the Tent family analysis only as retrospective sensitivity, not a cluster-robust, confirmatory, or independent-checkpoint win. This mixed setting is where routing can help: adaptation is useful in some cells, harmful in others, and the label-free evidence supports different actions.

The SAR row is an equally important negative control. Its KGA regret is 0.0016 while always-adapt is 0.0003; the certificate avoids the much worse always-freeze policy but does not beat the best fixed policy. The three candidate rows cannot be pooled into a claim that KGA always beats both. They show that routing utility depends on the adapter-induced benefit distribution and on how well the evidence predicts it.

Table 4: Canonical reconciled result panel. Lower regret is better. KGA, Adapt, and Freeze are regret to the per-condition oracle. FA_u is marginal false adaptation. Coverage is $\mathbb{P}(\text{ADAPT or FREEZE})$, not adapt rate. Values are generated directly from the canonical JSON; 391983f4fa09d65c256ab8c620fea772866b652c219dad0c36574ce38c740482 is the SHA-256 of its compact-source manifest. The table reports point estimates; this compact paper does not promote a row to a CI-robust win unless an inference artifact at the declared unit is present. Rows are protocol-specific and must not be pooled across datasets, adapters, or metrics. “Office-Home replication” denotes the archived 54-cell test-stream-seed replay, not the later five-independent-checkpoint audit in Table 6. For PACS, $n = 12$ counts equal-weight domain-seed inference units, while FA_u and coverage summarize 216 repeated decision evaluations (108 paired settings). The iWildCam row is withheld because the archived scorer used sklearn macro-F1 rather than the official WILDS label-present macro-F1 contract; its stored values are audit history, not canonical numerical evidence. The ImageNet-C SAR row uses a learning rate $16\times$ above SAR’s published setting (4×10^{-3} versus 2.5×10^{-4}) and leaves the final ResNet stage adaptable; no control at SAR’s official operating point has been run.

Panel	n	KGA	Adapt	Freeze	FA_u	Coverage
Office-Home M-v2	35	0.0158	0.0468	0.0158	0.0000	0.3143
Office-Home replication	54	0.0215	0.0458	0.0217	0.0000	0.2778
iWildCam H-v2		withheld: official-metric rerun required				
ImageNet-C SAR	135	0.0289	0.0529	0.0319	0.0074	0.2074
ImageNet-C TENT	135	0.0145	0.0191	0.0145	0.0000	0.0000
ImageNet-C EATA	135	0.0007	0.0001	0.0342	0.0074	0.8963
PACS	12	0.0431	0.0176	0.0446	0.0093	0.8148
ImageNet-R	480	0.0150	0.0064	0.0325	0.0000	0.4042
CIFAR-10-C TENT	2160	0.0016	0.0080	0.1239	0.0000	0.6787
CIFAR-10-C EATA	2160	0.0013	0.0033	0.1313	0.0000	0.6356
CIFAR-10-C SAR	2160	0.0016	0.0003	0.1405	0.0000	0.6694
Camelyon17 OOD	18	0.0000	0.0000	0.1381	0.0000	1.0000
Camelyon17 B-v2 TENT	108	0.0296	0.0097	0.0820	0.0093	0.3704
Camelyon17 B-v2 EATA	108	0.0083	0.0040	0.0911	0.0000	0.5648
Camelyon17 B-v2 SAR	108	0.0006	0.0016	0.1001	0.0000	0.6574
RxRx1 J	60	0.0000	0.2531	0.0000	0.0000	1.0000
CIFAR-10.1 K	48	0.0017	0.0190	0.0017	0.0000	0.4583

8.1.2 Protocol-Matched POEM and AETTA Comparison

The archived head-to-head on the same locked CIFAR-10-C stress stream reports KGA regret to oracle of 0.0016, below the POEM-style result (0.0088) and the AETTA-style result (0.0073). Using the paper’s baseline-minus-KGA convention, its ordinary paired-bootstrap 95% intervals are $[0.0057, 0.0088]$ for POEM and $[0.0044, 0.0071]$ for AETTA; only the paired p -values were Holm-adjusted. The artifact used an earlier KGA action vector, so synchronization with the current exact-rank policy is pending. These are historical protocol-matched ports, not a current-policy comparison or a claim about untouched official implementations.

These controlled results use empirical calibration. Leaving one cell out prevents that cell from directly fitting its own prediction or radius. It does not establish exact split-conformal coverage when corruption cells are dependent. Thus zero observed false adaptations describes this protocol; it does not guarantee the same result for every future corruption family.

Table 5: Retrospective current-policy sensitivity over six observed CIFAR-10-C corruption families. Each cell reports baseline regret minus KGA regret and an ordinary, unadjusted family-bootstrap 95% interval; positive values favor KGA. Holm p -values use the preregistered six candidate-by-baseline sign-flip family. The within-Tent two-contrast Holm value is 0.03125 but is post hoc. This table is conditional on one archived checkpoint and does not provide simultaneous intervals, independent-checkpoint inference, prospective confirmation, natural-shift evidence, or official POEM/AETTA superiority.

Candidate	Adapt gap [95% CI]	Freeze gap [95% CI]	Six-way Holm p (A/F)
Tent	0.00633 [0.00254, 0.00954]	0.12229 [0.06331, 0.17772]	0.09375/0.09375
EATA	0.00195 [-0.00052, 0.00438]	0.13001 [0.06977, 0.18754]	0.25000/0.09375
SAR	-0.00122 [-0.00267, -0.00007]	0.13891 [0.08031, 0.19686]	0.90625/0.09375

Table 6: Office-Home protocols are different experiments, not conflicting versions of one row. Entries are accuracy regret to the per-condition oracle; lower is better. The first row is the canonical single-candidate KGA replay with fixed SAR. The focused route is invalid, so only the post-hoc fixed-SAR opportunity regret (.0027) and freeze regret (.0230) are retained; its actions have no routing or certificate interpretation. A/F/U are adapt/freeze/abstain actions.

Protocol	n	Router / Adapt / Freeze	A/F/U
Archived stream-seed replay	54	.0215 / .0458 / .0217	1/14/39
Five-checkpoint candidate audit	15	invalid / .0027 / .0230	n/a

8.2 Large-Scale Corruption: ImageNet-C

ImageNet-C is candidate-dependent. SAR has KGA/adapt/freeze regrets 0.0289/0.0529/0.0319, a pooled point edge over both fixed policies, but with one false adaptation among 135 cells and without a promoted seed-robust beats-both claim. Tent ties the always-freeze point estimate and makes no strict decisions in the reconciled row. EATA nearly tracks always-adapt but has a larger KGA regret (0.0007 versus 0.0001). These results reject a dataset-level label such as “ImageNet-C win”: the correct unit is the candidate-specific policy. This SAR result uses a learning rate $16\times$ higher than SAR’s published setting (4×10^{-3} versus 2.5×10^{-4}), with the final ResNet stage left adaptable—an aggressive configuration at which SAR is expected to be fragile. No control at SAR’s official operating point has been run; this is a claim about this configuration, not about SAR as published.

8.3 Natural and Semantic Shifts

8.3.1 Office-Home

The primary Office-Home row is conservative. KGA regret is 0.0158, tying always-freeze (0.0158) and below always-adapt (0.0468), with 0 adapt decisions. This is a descriptive point-estimate no-harm tie, not a routing win. The archived protocol does not supply a predeclared uniform stability guarantee for the full-fit versus out-of-fold benefit predictor.

The 54-cell Office-Home stream-seed replication in Table 4 has KGA regret 0.0215 versus 0.0458 and 0.0217, a small point edge over both fixed policies. The three run seeds do not record independent checkpoint identities, and the seed-level interval includes zero, so this is not a robust win.

The post-hoc five-checkpoint Office-Home study is a candidate opportunity audit, not a valid KGA result. It uses five independently trained source checkpoints, the three target-test domains, one fixed IID/tiny condition, and Tent, EATA, and SAR. Checkpoint identity passed, but routing and serialization did not. The identifying agreement relation is binary-only and does not apply to the 65-class task; the old spectral estimate was sign-indeterminate and unbounded. Thirteen of 15 anchors were negative, 9 of 60 stored $\hat{b}$ values fell outside $[0, 1]$, and EATA and SAR produced exactly the same predictions in 10 of 15 cells. Each checkpoint has only three LOO cells, whereas a 90% exact-rank single-candidate radius requires at least 10 total cells. All five result files fail strict JSON and contain 57 literal `Infinity` values in total.

The archived exploratory implementation printed ABSTAIN in all 15 cells, but those outputs have no safety, routing, or certificate meaning. The valid descriptive quantities are checkpoint accuracies and candidate opportunity: freeze is 0.5711, SAR is 0.5914, their paired difference is 0.0203 (95% t -interval $[0.0123, 0.0283]$), and the per-cell oracle is 0.5941. Oracle minus SAR is 0.0027 (95% interval $[-0.0024, 0.0077]$). The best adapted candidate improves over freeze in 13 cells, ties in one, and harms in one. Because the target was already opened and SAR was chosen after comparing the audit means, these statistics are post-hoc opportunity evidence only.

8.3.2 Camelyon17

The archived Camelyon17 OOD evaluation yields zero KGA and always-adapt regret against 0.1381 for freeze. All 18 cells are helpful, so this already-opened panel is a descriptive always-adapt one-sided diagnostic rather than prospective evidence of selective routing. Within-seed candidate diagnostics remain separate from this OOD endpoint.

8.3.3 RxRx1

The displayed RxRx1 row is model seed 0: KGA and freeze regret are zero against 0.2531 for always-adapt, with no adapt decisions. A separate audit across three independently trained model checkpoints repeats the same all-freeze tie. It strengthens model-variation reproducibility but adds no adapt exposure. RxRx1 is therefore a descriptive no-harm case, not a routing win.

8.3.4 iWildCam

iWildCam is not assigned a canonical numerical row. The archived scorer used sklearn macro-F1, which averages prediction-only classes, rather than the official WILDS metric, which averages labels present in the target truth. The archived runtime, sample identities, and target population were not sealed under the current contract, so its diagnostic scores and routing actions remain audit-only. A pinned rerun with the official metric, sealed sample identifiers, and a population hash is required before promotion. The older beats-both flag and the later replay under the same invalid metric contract remain hash-locked only as superseded audit history.

8.3.5 Locked CCT-20 Camera-Location Shift

We evaluated the frozen KGA rule on Caltech Camera Traps 20 (CCT-20), a camera-location shift benchmark [55]. The study crossed 5 independently trained checkpoint tensors with 9 target camera locations. This produced 45 checkpoint–location cells, but the exact test uses only the nine camera locations as clusters. The checkpoints are repeated models, not five independent studies, and the 45 cells are not 45 independent environments.

Table 7: Locked CCT-20 primary contrasts. The estimand is baseline regret minus KGA regret, so positive values favor KGA. Intervals use a paired two-way product bootstrap that resamples checkpoint rows and location columns independently. Two 97.5% per-comparison intervals form a 95% Bonferroni family. Exact p -values are one-sided sign-flip tests over the nine location means; Holm adjusts only these p -values.

Comparator	Baseline regret – KGA regret	Bonferroni 97.5% CI	Exact p	Holm p	Reject at .05
Always adapt	0.18190227	[0.08501071, 0.31066374]	0.00195312	0.00390625	yes
Always freeze	0	[0, 0]	1	1	no

The locked target contained 23275 images: 6950 probe images and 16325 evaluation images. The primary score is set-membership top-1 accuracy: a prediction is correct when it matches any distinct archived category for the image. This is not an official single-label CCT-20 leaderboard reproduction. We used the official Tent implementation at commit [e9e926a](#), with one non-episodic Adam step at learning rate 10^{-3} and a state reset for every checkpoint–location cell.

Provenance ledger. Cell-level outcomes remained unopened before target execution. Aggregate target label metadata had already been inspected during candidate ranking, so this is not a label-unopened study. Item-level target labels were deserialized once, after the one-shot scoring marker was spent. The release chain also records the numeric-v2 repair in a sealed addendum created before outcome access. That repair replaced an unstable divergence calculation with a stable Gaussian KL computation; it did not change the model, data split, or scoring rule. We treat it as protocol provenance, not result-driven tuning. All primary outcome values and action/effect counts come from the sealed, content-addressed, receipt-linked CCT-20 release manifest and its generated macros (canonicalized inference-document SHA-256 5d1ba74d9a8b16ef24746ca3d95969e635d34fd4112806f6bf91b4b45a6cf907).

Action ledger. Across the 45 cells, KGA issued 0 ADAPT, 44 FREEZE, and 1 ABSTAIN decisions. Strict decisions were therefore made in 44 of 45 cells. The false-adapt count was 0 of 45 cells, but this value is mechanical because the gate never selected ADAPT. It does not show that the gate can identify helpful adaptation.

The exact sign-flip interpretation is conditional on sign symmetry and exchangeability across the nine location means; it is not a distribution-free guarantee for unseen locations.

Primary result. KGA exactly matched always freeze. Its contrast against always freeze was 0 with Bonferroni 97.5% interval [0, 0], exact location sign-flip $p = 1$, and Holm $p = 1$. Against always adapt, the contrast was 0.18190227 with interval [0.08501071, 0.31066374], exact $p = 0.00195312$, and Holm $p = 0.00390625$. Adaptation benefit is adapted accuracy minus frozen accuracy: 1 cell was helpful, 0 were exactly zero, and 44 were harmful. These counts explain why freezing was useful here. They do not show that the gate learned when to adapt.

Safe-utility and claim ledger. The locked safe-utility check uses pointwise 95% evidence. Against always adapt, the point estimate was 0.18190227 with interval [0.09374776, 0.29138626]. Against always freeze, it was 0 with interval [0, 0], compared with the frozen noninferiority margin -0.005. The generated pass flag is yes. Thus KGA passed the protocol’s narrow safe-utility check by retaining frozen predictions and avoiding the loss from always adapting. This is not a general

safety claim. The strong-success claim is unsupported because KGA neither beat always freeze nor produced meaningful ADAPT exposure.

Release verdict. SAFE UTILITY ONLY. The prospective CCT-20 result passes only the locked safe-utility check; it does not establish the preregistered strong-success claim. In this ledger, “preregistered” means internally sealed and time-stamped before source training and target outcomes; it does not denote a public registry.

Limits and next step. All 55 development cells had negative Tent benefit, so the gate had no positive ADAPT example. The ridge fit used 10 checkpoint–location rows from only two camera units for 11 recorded features. Calibration used 45 rows collapsed to nine cis-test camera locations, while the locked targets came from trans-test locations. Coverage therefore needs an unverified cross-split exchangeability assumption. Target Mahalanobis diagnostics ranged from 7.12 to 485.67 (median 28.73), showing strong extrapolation; this diagnostic never changed an action and is not a formal coverage guarantee. One natural dataset also cannot establish a general natural-shift win. The next study should retain the same outcome-blind release path, add diverse development units, require useful exposure to both ADAPT and FREEZE, and repeat the test on a second locked natural-shift dataset. Cell-level 16-indicator macro-F1 is archived as descriptive secondary evidence; no post-hoc aggregate or inference claim is made.

8.4 Negative and Diagnostic Tracks

8.4.1 PACS

PACS KGA regret 0.0431 exceeds always-adapt’s 0.0176. Its archived aggregate lacks the residual detail needed for full gate replay, so it remains a negative aggregate diagnostic rather than a complete certificate evaluation.

8.4.2 ImageNet-R Backbone Breakdown

ImageNet-R KGA regret 0.0150 exceeds always-adapt’s 0.0064 across the ten-backbone architecture panel, and no backbone beats both fixed policies. KGA is worse than always-adapt on 8 of 10 backbones; four of those eight have a 0% harmful base rate, so routing has no harm to avoid. On `efficientnet_b0` (100% harmful) and `swin_t` (60.4% harmful), KGA improves over always-adapt but exactly ties always-freeze and makes no adapt decisions. The pattern is consistent with the regime map: routing can add value only where candidate harm is present, but these two rows are fail-closed no-harm ties rather than routing wins.

8.4.3 CIFAR-10.1

CIFAR-10.1 ties always-freeze at 0.0017 and is retained as a negative low-margin cross-seed diagnostic. None of these nulls proves structural non-identifiability; weak evidence, low margin, estimator inadequacy, or calibration transfer can produce the same pattern.

8.5 What the Panel Establishes

The panel supports neither universal improvement nor a clean, CI-robust beats-both win on a natural dataset. Its strongest positive routing evidence is controlled and candidate-specific. Its natural evidence is mainly no-harm, often obtained by declining adaptation. CCT-20 adds a locked

Table 8: Claim discipline for the reconciled panel.

Observation	Supported interpretation
CIFAR-10-C Tent result	exact-rank point gain; retrospective current-policy six-family intervals positive; preregistered six-comparison Holm fails
CIFAR-10-C EATA result	point-estimate gain; current adapt-side family interval includes zero; preregistered six-comparison Holm fails
CIFAR-10-C SAR loss	candidate dependence is material
Office/Rx no-harm	descriptive fail-closed behavior; little or no adaptation exposure
CCT-20 locked result	safe-utility check passes; KGA ties freeze, rejects always adapt, and has zero ADAPT exposure
iWildCam	numerical claim withheld pending official-metric, population-sealed rerun
Camelyon17 OOD tie to adapt	archived opened one-sided diagnostic; no selective-routing test
PACS/ImageNet-R losses	completed negative diagnostics, retained in scope
Empirical abstention	uncertainty under the fitted pipeline, not proof of structural unknowability

safe-utility result: KGA rejects harmful adaptation but does not select the single helpful cell. Together, these findings support fail-closed and rollback mechanics in the observed protocols, but they do not establish an adapt-side safety property or broad accuracy improvement.

9 Sensitivity, Failure Analysis, and Limitations

9.1 Radius and Decision Yield

Increasing ε weakly reduces strict decisions and can increase regret when abstention retains a suboptimal frozen model. Removing the radius increases yield but removes the coverage-based reason for a strict sign. Consequently regret and certification are separate axes; a low-regret point gate is not automatically a valid gate.

9.2 Safety Level and Calibration Size

Smaller α raises the order-statistic rank and generally widens the interval. With a small calibration set, the requested rank may exceed the available residuals, in which case the defensible finite-sample response is an infinite radius and abstention. Clamping to the largest residual gives a finite number but does not preserve the same nominal split-conformal statement in that edge case.

9.3 Estimator Quality

A weak h_θ can make the radius large or shrink $\hat{\Delta}$ toward zero, causing avoidable abstention. A flexible model can overfit and make in-sample residuals optimistic. Out-of-fold prediction reduces direct leakage but does not remove dependence among related conditions. These effects must be diagnosed with residuals at the intended deployment unit.

9.4 Adapter and Configuration Transfer

Changing the adapter changes both Z and Δ . A benefit model calibrated on Tent therefore has no automatic certificate for EATA or SAR. The maintained implementation refits the benefit model and radius per dataset and adapter. A checkpoint, optimizer, parameter subset, or update-step change is also a calibration-contract change.

9.5 Domain Transfer and Exchangeability

Worst-group development radii can be useful sensitivity analyses, but they protect a novel target group only under an additional dominance or exchangeability condition. Taking a maximum over observed development groups does not by itself certify an unseen group. When that condition is not defensible, the system should fail closed and the paper should report the row as descriptive.

9.6 Population Budget Is External

The frontier is exact relative to $\mathcal{C}_\beta$, but an unlabeled deployment batch cannot verify $|\gamma| \leq \beta$. A credible budget must come from domain knowledge, a declared coupling, or historical labeled deployments under a transfer assumption. Setting $\beta = 0$ assumes zero drift; it is not a safe default.

We tested a source-development benefit-scale proxy as a possible operational surrogate for β ; it does not estimate the theorem’s calibration-drift bound. On CIFAR-10-C, the proxy is 1.4–50 $\times$ too small relative to evaluation-scale discrepancies: 24–73% of cells violate the putative budget, and commit error is 0.4–16.6%. On ImageNet-C, a larger proxy often removes most decisions; 5 of 10 configurations return zero commitments on all 405 cells. We therefore withdraw this surrogate route. The population β remains externally supplied from domain knowledge or a stated assumption.

The theory-to-algorithm link is direct but conditional. Both layers target Δ , as Eq. (6) shows. This shared target does not prove that Z is risk-aligned, that h_θ transfers, or that residual coverage holds in a new domain. Those remain external validity requirements. KGA applies the three-way decision through an interval; it does not learn the population assumptions.

9.7 Controlled-Grid Dependence and Natural-Shift Limits

CIFAR-10-C is large and mixed, but related cells share corruption families, severities, and checkpoints. Leave-one-cell-out evaluation is weaker than holding out an entire corruption family or an independent physical deployment. We therefore describe Tent and EATA as controlled exact-rank point gains. The retrospective current-policy family analysis gives Tent positive ordinary intervals, but the preregistered six-comparison Holm gate fails and the six families share one archived checkpoint. The older earlier-policy cluster artifact remains separate historical evidence. Neither result proves independent-checkpoint stability or natural-domain transfer.

Natural evidence is mostly no-harm. Several natural rows tie the better fixed policy because the candidate is one-sided and KGA makes few or no adapt decisions. Such behavior can be appropriate, but it does not demonstrate an accuracy breakthrough. A genuinely unopened or hidden-label natural beats-both result remains open.

9.8 Search, Multiplicity, and Selective Reporting

Canonicalization does not create missing evidence. The reconciler removes numerical contradictions and preserves source hashes, but it cannot recover unlogged residuals, pre-registration, or independent replication. PACS remains aggregate-only; Office-Home retains a transfer-stability caveat; iWildCam is withheld for a metric-contract rerun; ImageNet-R is an architecture-panel diagnostic. The table exposes these limitations rather than imputing missing values.

Search history. The repository contains many exploratory configurations. One superseded review census reported 1,387 determinations and 326 positives (23.5%); a broader archived scan reported 1,427 determinations and 345 positives. Because neither census froze one reproducible family definition, file list, and commit, neither total is promoted as the release multiplicity family. We instead report a current post-hoc reconciled panel and do not relabel its favorable rows as confirmatory discoveries. Future claims should declare the candidate, calibration unit, target unit, seed structure, inference unit, and beats-both rule before test labels are opened.

9.9 Physical Deployment and Runtime Limits

The repository contains a dashboard and a camera-study protocol, but no held-out physical result. We therefore make no real-camera claim. A future study should separate sessions by day and device, keep fitting and calibration separate from testing, hide live labels, replay the same stream for every policy, and report accuracy, regret, false adaptations, decision rates, and total latency.

10 Reproducibility and Artifact Scope

10.1 Canonical Numerical Source of Truth

The point estimates in Table 4 have one numerical source: `experiments/kbound/results/reconciled_panels_v1/canonical_panel_results.json`. Its compact source manifest has SHA-256 391983f4fa09d65c256ab8c620fea772866b652c219dad0c36574ce38c740482. Named auxiliary artifacts support only their stated roles. The code-bound current-policy family sensitivity is retrospective and fails the preregistered six-comparison Holm gate; the earlier Tent cluster artifact and archived POEM/AETTA comparison remain historical. None overwrites the canonical point-estimate panel. The generated result manifest indexes every promoted claim and its scope.

10.2 Reproduction Pipeline

From the repository root, the maintained sequence is:

```
BUILD_LONG_TMLR=1 bash docs/research/kbound/scripts/build_pdfs.sh
```

The script reconciles source-hashed records, synchronizes structured manifests, regenerates tables, macros, and figures, and builds both layouts. It consumes frozen model outputs; it does not rerun training.

10.3 Calibration Identity and Implementation Contract

The maintained API binds a certificate to the checkpoint, adapter configuration, evidence schema, benefit estimator, residual construction, safety level, and scoring unit. The deployment path must fail closed when any identity differs. `KGA.certify_evidence` requires a frozen estimator, the exact schema, a protocol hash, and a separate residual set. The lower-level `certify` function is an audit interface; it does not construct evidence or validate transfer by itself.

10.4 Formalization and Lean Scope

Lean artifacts and validators check the encoded theorem cores and selected finite constructions. They do not certify dataset correctness, preprocessing, empirical transfer, exchangeability, or every deployment assumption stated in prose. Machine checking strengthens the statements it encodes; it does not validate the benchmark pipeline outside those statements.

10.5 Release Checklist and Audit Logs

The Phase-1 provenance audit matches all 106 canonical source/compact hash pairs, all 10 recoverable configuration hashes, and all 6 archived checkpoint hashes; it also records a 21-file current-code snapshot. Historical runs that did not serialize full identity remain explicitly partial rather than being backfilled. The release gate runs reconciliation tests, claim-ledger and result-manifest validation, manuscript checks, LaTeX builds, checksum verification, and visual PDF inspection. Build logs, the canonical manifest, the provenance seal, and the final PDFs are archived together.

11 Conclusion

11.1 Main Conclusion

K-Bound treats test-time adaptation as a decision about a proposed update. In the binary population model, a declared drift class supports a strict adapt or freeze action exactly when $|M| > \beta$. Inside $|M| \leq \beta$, the available label-free evidence does not support a uniform strict direction, so abstention is required. KGA handles the practical finite-sample problem. It estimates the same benefit Δ and acts only when the calibrated interval excludes zero. The two layers share a target, but their uncertainty quantities are not interchangeable: M is not $\hat{\Delta}$, and β is not ε .

The evidence supports a specific conclusion. In controlled mixed conditions, KGA gives exact-rank point gains over both fixed policies for Tent and EATA, while SAR loses to always-adapt. Tent’s retrospective current-policy six-family intervals are positive, but its preregistered six-comparison Holm values are 0.09375 against both baselines. No cluster-robust or confirmatory gain is therefore promoted. On natural and semantic shifts, KGA mainly matches the better one-sided policy or records a negative diagnostic. In the locked CCT-20 study, it matched always freeze and beat always adapt, but made no ADAPT decisions and therefore failed the stronger protocol test. These results support a fail-closed rollback design for avoiding unsupported updates in the observed protocols. They do not establish general safety, selective natural-shift routing, or a universal accuracy claim.

11.2 Next Confirmatory Experiment

The next empirical step is clear. A new prospectively locked study should lock the adapter, evidence schema, calibration unit, scoring unit, and inference unit before target labels are opened. It should use a separate calibration partition and then evaluate unopened natural target environments containing both helpful and harmful updates. A meaningful test requires nontrivial adapt and freeze exposure, predeclared paired regret gaps against both fixed policies, and labels joined only after every decision is fixed. Such a study would answer the practical question at the center of this paper: can label-free evidence identify when an update is worth the risk?

A Compact Assumption and Claim Ledger

B Calibration and Evaluation Contracts

Controlled-grid contract. The fitting unit is the condition cell. An out-of-fold benefit estimate is produced without fitting on the scored cell, and the residual radius for that cell excludes its residual. This is leave-one-condition-out cross-fitted empirical residual calibration. Related cells may still be dependent through corruption family, checkpoint, or seed. Exact split-conformal wording is not used for this branch.

Table 9: Claim-to-support ledger for the compact submission. “Requirement” states what must be true in addition to the displayed result.

Claim	Type	Requirement	Evidence	Caveat
Interior matched-evidence impossibility	theorem	$\beta > 0$, $ M < \beta$, full fibre richness	Thm. 1	binary, 0/1 loss
Closed-band abstention	theorem	strict directional semantics and rich class	Prop. 1	zero benefit blocks a strict claim
Exact strict-commitment frontier	theorem	fixed augmented evidence, declared $\mathcal{C}_\beta$	Thm. 2	β externally supplied
Marginal false-adapt control	theorem	Eq. (7)	Thm. 4	does not bound FA_c
CIFAR-10-C Tent routing gain	empirical	frozen Tent controlled-grid protocol	Tables 4 and 5	exact-rank point gain; positive ordinary six-family intervals; preregistered Holm fails
CIFAR-10-C EATA point gain	empirical	frozen EATA controlled-grid protocol	Tables 4 and 5	adapt-side family interval includes zero; preregistered Holm fails
Natural no-harm rows	empirical	archived split and candidate remain fixed	Table 4	often zero adapt exposure
ImageNet-C candidate dependence	diagnostic	candidate rows interpreted separately	Table 4	no dataset-wide win
PACS/ImageNet-R/CIFAR-10.1 nulls	diagnostic	canonical replay scope retained	Table 4	null does not prove unknowability
Universal improvement	not claimed	would require broad held-out dominance	none	contradicted by negative rows
Real-camera validation	pending	locked sessions and fresh held-out recordings	protocol only	no result in this paper

Natural-shift contract. Development conditions fit the benefit model; a disjoint calibration partition supplies residuals; choices are frozen; and the target partition is scored once where the archived protocol records all three stages. When the saved evidence does not support that complete chain, the row is marked descriptive or diagnostic. Replaying a table does not strengthen its original design.

The candidate TTA step itself is transductive. Episodic candidates update on each unlabeled evaluation batch before predicting that batch. Online candidates update on a separate auxiliary stream, but prediction still uses evaluation-batch BatchNorm statistics. Thus “disjoint” above refers to development, residual-calibration, and target partitions (and to the recorded auxiliary stream versus evaluation sample identities); it does not mean that candidate inference is inductive. No target labels enter adaptation, prediction, evidence, or the action.

Action contract. Every action is computed from Z , $\hat{\Delta}$, and ε without target labels. Offline labels determine Δ , oracle action, regret, and false-commit indicators only after the action is fixed. Abstention resolves to the frozen prediction for task-risk accounting, while its log retains the distinction from a certified freeze.

Configuration contract. The checkpoint, candidate adapter, optimizer, adapted parameter subset, update count, evidence schema, feature normalization, benefit estimator, calibration split, α , and software version are part of one calibration identity. Changing any of them requires either a declared transfer argument or recalibration. The fail-closed default on identity mismatch is non-commitment.

Table 10: Action exposure in replayable canonical rows. Counts are not independent experiments: they are evaluation units within the protocol shown. A zero false-adapt count is informative only when adaptation was attempted and the residual pool did not make the inequality automatic. Counts were regenerated from source manifest SHA-256 391983f4fa09d65c256ab8c620fea772866b652c219dad0c36574ce38c740482.

Row	n	A	F	U	false A	FA _u	repetition / scope
Office-Home primary	35	0	11	24	0	.0000	2 test-stream seeds; ties freeze
Office-Home stream-seed repl.	54	1	14	39	0	.0000	3 test-stream seeds; point-only edge
ImageNet-C SAR	135	13	15	107	1	.0074	5 seeds $\times$ 27 cells
ImageNet-C Tent	135	0	0	135	0	.0000	5 seeds $\times$ 27 cells
ImageNet-C EATA	135	120	1	14	1	.0074	5 seeds $\times$ 27 cells
ImageNet-R	480	165	29	286	0	.0000	4 seeds $\times$ 10 backbones $\times$ 12 cells
CIFAR-10-C Tent	2160	1107	359	694	0	.0000	5 seeds $\times$ 432 cells
CIFAR-10-C EATA	2160	1241	132	787	0	.0000	5 seeds $\times$ 432 cells
CIFAR-10-C SAR	2160	1446	0	714	0	.0000	5 seeds $\times$ 432 cells
Camelyon17 OOD	18	18	0	0	0	.0000	archived opened helpful-only OOD subset
Camelyon17 B-v2 Tent	108	40	0	68	1	.0093	3 within-seed grids; diagnostic
Camelyon17 B-v2 EATA	108	61	0	47	0	.0000	3 within-seed grids; diagnostic
Camelyon17 B-v2 SAR	108	71	0	37	0	.0000	3 within-seed grids; diagnostic
RxRx1 J	60	0	60	0	0	.0000	displayed model seed 0; separate 3-checkpoint audit also all-freeze
CIFAR-10.1 K	48	0	22	26	0	.0000	locked cross-seed replay

C Decision Exposure and Repetition Audit

Point regret alone can conceal whether a no-harm result exercised the adapt branch. Table 10 therefore reconstructs action counts from the same canonical panel used in Table 4. A/F/U denote adapt, freeze, and abstain. PACS is omitted from the action table because the released aggregate cross-checks regret but does not contain the per-cell benefit estimates and residuals required for a complete gate replay. iWildCam is omitted because its archived actions were scored under the wrong macro-F1 contract.

The largest adapt exposure occurs on the controlled CIFAR-10-C rows. This is why those rows are the main empirical test of the adapt-side certificate. Primary Office-Home and RxRx1 have zero adapt exposure, while the 54-cell Office-Home stream-seed replication has one adapt decision. Their zero false-adapt values barely test whether the gate can distinguish a harmful adaptation from a helpful one; they show that the controller declines the candidate under those evidence/calibration contracts. The invalidated five-checkpoint Office-Home route and the withheld iWildCam actions are not counted. Camelyon17 OOD has the opposite limitation: every archived condition is helpful and every action adapts, so there is no harmful opportunity to reject. ImageNet-C EATA has substantial adapt exposure and one false adaptation, while Tent abstains everywhere. Those three patterns are operationally different even when their regrets are small.

The repetition column also prevents a common inflation of evidence. Five seeds of one checkpoint and grid protocol are not five independently designed studies. Ten ImageNet-R backbones are a panel of candidate policies, not ten target domains for one frozen gate. Three Camelyon17 within-seed grids permit a stability diagnostic but do not create an untouched hospital-domain replication. RxRx1 is stronger on model variation because the source checkpoint is independently trained three times, but its all-freeze behavior still provides no adapt exposure.

D Inference Discipline

D.1 Point Estimates, Paired Gaps, and Clusters

For a baseline $b \in \{\text{always-adapt}, \text{always-freeze}\}$, the paired routing gap at condition j is

$$G_{bj} = \text{regret}_{b,j} - \text{regret}_{\text{KGA},j}.$$

A positive mean gap favors KGA. All paired routing-gap intervals reported in this paper use this baseline-minus-KGA sign convention. A point-estimate beats-both statement requires both means to be positive. A CI-robust statement additionally requires the lower confidence limit for both paired means to exceed zero at the declared resampling unit. These definitions are intentionally separate: Table 4 can establish a point ordering, but not an inference claim without the associated unit and interval artifact.

Condition-level resampling treats cells as exchangeable and is optimistic when severities and compositions share a corruption family. Corruption-family resampling is more conservative but has few clusters. Seed resampling measures variation across repeated training or test streams only when the seeds represent the intended deployment randomness. The compact paper therefore does not use one generic “bootstrap CI” label. An inference claim must state whether the unit is cell, corruption-severity pair, corruption family, seed, backbone, domain, or physical session.

D.2 False-Adapt Accounting

The event in Eq. (2) is unconditional. Its empirical estimate divides false adapt decisions by all evaluated units, including freezes and abstentions. Conditional false-adapt divides by adapt decisions and is undefined when no adaptation occurs. A zero observed count should be accompanied by the number of adapt decisions; otherwise a controller that never adapts appears perfect. Table 10 exposes that denominator.

If the scored residual is included in the same finite residual vector whose order statistic decides the action, the number of residuals above the order statistic is algebraically constrained. Such coverage is not an independent empirical validation. Excluding the scored residual removes that direct identity but does not automatically provide the exchangeability or stability needed for a finite-sample theorem. Clean split calibration and LOO empirical calibration must therefore remain different protocol labels.

D.3 Multiplicity and Search History

An earlier repository census counted many exploratory beats-both fields, but it did not freeze the file list and commit needed to reproduce a current total. We therefore do not promote that count or its positive rate as release evidence. The panel in this paper is a post-hoc reconciled audit panel assembled from individually locked and historical protocols; it is not a globally preregistered confirmatory family. A result selected after a search cannot become confirmatory by correcting only the final positive rows. Future confirmatory families must name every candidate-track test before any test value is computed, and must retain failures and incomplete runs.

E Track-Level Provenance and Remaining Evidence

The provenance table is part of the result, not administrative metadata. A reviewer can trace a canonical row to compact records and their original archive hashes. If the original large artifact

Table 11: What is present in the compact release and what remains unavailable. “Replay” means the stored quantities suffice to recompute the displayed gate output under the released scorer.

Track	Released status	Present evidence	Remaining limitation
CIFAR-10-C	replayable	per-condition Δ , $\hat{\Delta}$, radius, action; 5 seeds; 3 candidates	related grid cells; no untouched corruption-family test in canonical table
ImageNet-C	replayable	per-condition records for 5 seeds and Tent/EATA/SAR	27-cell controlled configuration; candidate-specific, not one policy
Office-Home	replayable	canonical primary and 54-cell stream-seed records; later five-checkpoint audit reported separately	full-fit-versus-OOF transfer-stability premise not predeclared
iWildCam	withheld	archived opened test-split records; non-promoted official-metric diagnostic	wrong archived metric; population was not sealed and overlap is unauditale
Camelyon17	replayable diagnostic	archived opened OOD row plus separate B-v2 diagnostics	OOD row is one-sided and non-prospective; B-v2 is not untouched-domain evidence
RxRx1	replayable	displayed model-seed-0 row plus three model-checkpoint records and source hashes	all three checkpoints freeze throughout; no helpful test conditions
CCT-20	sealed complete	5 independent checkpoints, 9 target locations, actions, predictions, score, and inference hashes	only two fit locations; no ADAPT exposure; trans-target extrapolation
PACS	partial	three-seed aggregate and source cross-checks	missing per-cell $\hat{\Delta}$ and residuals prevent full gate replay
ImageNet-R	replayable	4 seeds, 10 candidate backbones, 12 conditions each	architecture aggregate is diagnostic, not one deployable router
CIFAR-10.1	replayable	locked cross-seed condition records	negative low-margin result; no natural beats-both claim
Physical camera	protocol only	dashboard, logging schema, planned session split	no fresh held-out physical recordings or result table

is not shipped, the manifest still records its byte count and digest. This guards against silent replacement of a result while allowing the release to omit raw datasets and oversized transient checkpoints.

Three upgrades would materially strengthen the empirical case. First, hold out entire shift families rather than individual controlled cells and fit all hyperparameters before opening the family. Second, add independent model and test-session seeds to the natural tracks that currently have one target replay. Third, run the physical protocol with days, devices, and objects assigned at the session level, then score every policy on the same locked streams. These upgrades address dependence and transfer; simply adding more rows from the same grid does not.

E.1 CCT-20 Location and Release Ledger

Table 12 reports all nine location clusters. Evaluation n is the number of evaluation images per checkpoint at that location. Each A/F/U triplet and each helpful/zero/harmful triplet therefore has denominator five, one cell for each checkpoint. Accuracy differences use one sign throughout: adapted minus frozen for adaptation benefit, and KGA minus the named fixed policy for the two policy columns.

The release ledger contains all 45 cells: 0 ADAPT, 44 FREEZE, and 1 ABSTAIN actions; 1 helpful, 0 zero, and 44 harmful adaptation effects. The release verdict remains SAFE UTILITY ONLY. The location table is descriptive; exact inference uses the nine location clusters, and no post-hoc aggregate is claimed for the secondary macro-F1 outcome.

Table 12: Complete locked CCT-20 location ledger. A/F/U denotes ADAPT/FREEZE/ABSTAIN. Within each location, the three effect columns are equal-weight means over five checkpoint cells. Those cells are repeated-model observations; the inferential cluster unit is the camera location.

Camera	Eval. n /checkpoint	A/F/U (5 cells)	Helpful/zero/harmful (5 cells)	Adapt acc. — freeze acc.	KGA acc. — adapt acc.	KGA acc. — freeze acc.
0	1302	0/5/0	0/0/5	-0.54178187	0.54178187	0
7	1283	0/5/0	1/0/4	-0.02930631	0.02930631	0
28	911	0/5/0	0/0/5	-0.05620198	0.05620198	0
40	800	0/5/0	0/0/5	-0.12325	0.12325	0
46	4432	0/4/1	0/0/5	-0.07233755	0.07233755	0
78	1633	0/5/0	0/0/5	-0.08389467	0.08389467	0
100	3441	0/5/0	0/0/5	-0.20348736	0.20348736	0
105	1540	0/5/0	0/0/5	-0.28922078	0.28922078	0
130	983	0/5/0	0/0/5	-0.23763988	0.23763988	0

F Extended KGA Implementation Contract

F.1 State Isolation

The frozen and candidate predictors must not share mutable normalization buffers, optimizer state, or parameter storage during preview. The candidate is created from a checkpoint, adapted on the unlabeled window, and evaluated without overwriting the active state. Only an adapt decision may replace the active checkpoint. Freeze and abstain discard the temporary state. A unit test should hash the active parameters before and after each non-adapt decision and require equality.

F.2 Evidence Determinism

Evidence extraction must define preprocessing, batch order, numerical precision, class ordering, normalization constants, and missing-value behavior. Paired features require frozen and candidate outputs on the same samples in the same order. Source statistics and ATC thresholds are immutable calibration artifacts. Evidence schema versions should be explicit rather than inferred from vector length, because two schemas can share a dimension while changing semantics.

F.3 Calibration Identity

The calibration identity should hash the source checkpoint, candidate adapter class, adapted parameter names, optimizer and learning rate, update steps, evidence schema, benefit-model hyperparameters, feature scaler, residual construction, α , and inference-unit definition. The radius belongs to that identity. Reusing it after a candidate or evidence change is not a benign software optimization; it changes the random variable whose residual is being covered.

F.4 Runtime Accounting

A deployment report should time capture/preprocessing, frozen inference, candidate adaptation, candidate inference, evidence extraction, benefit prediction, interval decision, copy creation, and rollback separately. Mean and tail latency should be reported at the decision-window level. Peak memory should include both model states and optimizer buffers. The compact manuscript leaves these cells unfilled because the canonical cross-dataset manifest does not provide one comparable runtime measurement. Omitting an unsupported number is preferable to quoting controller-only latency as end-to-end cost.

Table 13: Feature families used by the maintained implementation. All quantities are aggregate statistics of unlabeled inputs, frozen/candidate outputs, or the proposed update.

Family	Representative coordinates	Required state	Failure handling
Predictive uncertainty	mean entropy, confidence, quantiles	frozen/candidate soft-max	reject NaN or malformed probabilities
Marginal balance	normalized class-marginal entropy and change	predicted-class marginals	fixed class count and normalization
Model change	entropy gap, confidence drop, prediction disagreement	paired frozen/candidate outputs	require aligned sample order
Collapse signal	fraction above adapted confidence 0.9	candidate probabilities	threshold fixed before scoring
Distribution shift	free-energy shift, batch/running-statistic divergence	frozen logits and source statistics	missing source statistic forces non-commitment
Update magnitude	parameter-update ℓ_2 norm	shadow candidate delta	configuration hash must match calibration
Source-calibrated proxy	ATC-style threshold estimate	source-fixed threshold	target labels never used

Table 14: Operational failures and conservative responses.

Failure	Detection	Response
Checkpoint/adaptor mismatch	stored hash mismatch	rollback and recalibrate
Evidence outside support	distance or support audit	abstain
Missing/invalid feature	schema and finite-value check	abstain or freeze
Insufficient calibration size	rank $k > n$	infinite radius
Batch too small	minimum-size contract	wait or freeze
Residual transfer concern	group/residual diagnostic	widen or withhold claim
Continual-state drift	state and stream monitor	restore last certified state

G Evidence Inventory

The base panel has 11 coordinates; the canonical Camelyon17 rich-Z artifact has 17 and the Office-Home artifact has 18. Other natural tracks retain the 11-coordinate base panel. These protocol-specific dimensions are part of the fitted estimator contract. They should not be described as universal sufficient statistics: a different shift may hide the benefit direction from all listed features. Feature selection can improve empirical prediction, but cannot defeat the label-kernel construction of Lemma 1 over an unrestricted target fibre.

H Failure-Safe Deployment Checklist

The audit log should store a timestamp, dataset or stream identifier, checkpoint and adaptor hashes, evidence schema version, evidence vector, benefit estimate, radius, interval endpoints, action, rollback outcome, latency components, and configuration hash. Labels, if later revealed, are joined in a separate offline audit table. This separation makes it possible to verify that the live decision was target-label-free.

I Prospective Natural-Shift Confirmation Protocol

The sealed CCT-20 study narrows, but does not close, the empirical gap that matters most. Its 45 checkpoint-location cells contain 1 helpful and 44 harmful candidate updates, but KGA makes 44 FREEZE, 1 ABSTAIN, and 0 ADAPT decisions. It therefore matches always-freeze and improves on always-adapt: evidence of conservative harm avoidance under natural shift, not evidence of bidirectional routing or improvement over both fixed policies. The following protocol describes

the stronger second study still needed. It is a design for future evidence and is not counted as a completed experiment in this paper.

I.1 Lock Before Target Labels

The protocol should name one source checkpoint, one candidate adapter, one evidence schema, one benefit-model class, one calibration rule, one α , and one target-domain family. Adapter hyperparameters must be selected on development domains and then frozen. The lock should hash the checkpoint, code revision, configuration, calibration records, and planned scoring script. It must also name the test unit and the multiplicity family. A later change creates a new protocol version; it does not overwrite the failed or incomplete arm.

Natural heterogeneity should be built through independently defined environments rather than by pooling whichever benchmark conditions produce opposite signs. Examples include hospitals, camera locations, laboratory batches, acquisition days, physical devices, or visual domains. Environment membership may be observed at deployment, but the realized benefit label remains hidden. The target test environments should be absent from estimator fitting, residual calibration, threshold tuning, and candidate selection.

I.2 Three-Way Data Separation

Development data fit h_θ and select the evidence pipeline. A separate calibration set produces residuals for Eq. (8). The held-out target set is opened exactly once after all software and configuration hashes match the lock. Splitting adjacent frames, patches, or samples from the same session across these stages is leakage even when filenames differ. The split must occur at the environment or session level that defines deployment exchangeability.

The target stream is replayed identically for always-freeze, always-adapt, simple gates, KGA without a radius, and the full certificate. Model randomness should either be frozen for paired replay or repeated through independently trained checkpoints declared in advance. Every policy receives the same ordered windows. Labels are stored behind a separate audit interface and are unavailable to the live decision process.

I.3 Endpoints and Decision Rule

The primary endpoint is paired regret against each fixed policy. Success requires both baseline minus KGA regret gaps to be positive and both confidence intervals to exclude zero under the predeclared resampling unit. The false-adapt endpoint is FA_u with the adapt count reported beside it. Secondary endpoints are FA_c , adapt rate, freeze rate, abstain rate, decision coverage, task accuracy, and latency. Conditional false-adapt is descriptive unless a separate selective-risk result is proved.

The primary certificate rule remains unchanged: adapt if $\hat{\Delta} - \varepsilon > 0$, freeze if $\hat{\Delta} + \varepsilon < 0$, and abstain otherwise. The experiment must not lower ε , change α , or select a different benefit regressor after observing target labels. Exploratory sensitivity analyses may be reported, but they remain outside the primary bar.

I.4 Interpreting Outcomes

Four outcomes are scientifically useful. A beats-both result with nontrivial adapt and freeze exposure supports routing utility under the locked natural-shift class. A tie to the better fixed policy with few errors supports no-harm behavior but should not be called dominance. High abstention with

Table 15: Minimum lock for a confirmatory natural-shift KGA experiment.

Component	Must be fixed	Audit evidence
Model and candidate	checkpoint, adapter, optimizer, steps, adapted parameters	hashes and immutable configuration
Evidence and estimator	schema, normalization, model class, hyperparameters, seed	feature contract and fitted-artifact hash
Calibration	split unit, residual construction, exact rank, α	calibration IDs, residuals, radius, no target overlap
Target evaluation	environments, windowing, order, exclusion rules, transductive/inductive candidate semantics	sealed manifest and one-time scoring log
Primary inference	two paired regret gaps, cluster unit, interval method	code and random seed fixed before label reveal
Safety	FA _u and adapt exposure	action log joined to labels only offline
Multiplicity	all candidate-track arms in family	time-stamped inventory including failures
Runtime	end-to-end components and peak memory	raw timing rows on the target hardware

Table 16: Dependency map. A check in one row does not validate premises in another row.

Statement	Inputs	Conclusion	Not concluded
Disagreement reduction	binary labels, 0/1 loss, $\mu_T(D) > 0$	$\Delta = 2\mu_T(D)(M + \gamma)$	calibration transfer or observability of γ
Interior impossibility	label-free fibre, $\beta > 0$, $ M < \beta$, class richness	evidence-identical laws with opposite nonzero benefits	every empirical abstention is structural
Closed-band result	strict directional semantics and boundary law	no uniformly sound strict action on $ M \leq \beta$	adapt/freeze have different task risk at $\Delta = 0$
Frontier sufficiency	declared $ \gamma \leq \beta$ and fixed M	direction is fixed when $ M > \beta$	label-free estimation of β
Finite-sample certificate	marginal interval coverage	marginal false-adapt/freeze control	conditional error control or risk alignment
KGA benchmark	fitted h_θ , archived radius, frozen protocol	measured regret and actions on saved units	universal transfer to a new domain
Lean build	encoded definitions and assumptions	kernel-checked formal statements	correctness of unencoded data and deployment assumptions

low error indicates a conservative certificate and motivates better evidence or more calibration data. A loss or excess false adaptation falsifies the tested transfer/calibration contract and must remain in the result family. None of these outcomes alone establishes the population band: that conclusion still requires the declared class and its structural assumptions.

J Reviewer Guide to Theorem Dependencies and Nonclaims

The compact paper uses three nested logical statements. First, the disagreement reduction is an identity under the binary setup. Second, the population frontier is a uniform statement over a rich, declared class. Third, the empirical certificate is a marginal implication from interval coverage. The statements share Δ but neither the assumptions nor the quantifiers are interchangeable.

Why the boundary is separate. For $|M| < \beta$, the construction chooses two latent drifts that yield opposite nonzero benefits. At $|M| = \beta > 0$, one admissible drift can make the benefit zero while another retains the sign of M ; opposite nonzero signs need not exist. Both cases block a uniform strict commitment, but for different reasons. At $M = \beta = 0$, every law in the zero-drift class has zero benefit under the model. Abstention remains the maximal action only under the paper’s strict-direction convention, not because either task-risk choice is worse.

Why coverage is the premise. The certificate proof is elementary once Eq. (7) holds: on the coverage event, an interval wholly above zero cannot contain a nonpositive Δ . The scientific burden is therefore to justify coverage for the calibration/deployment pair. Exchangeability is one route; known shift correction or another valid construction may be another. Risk alignment can explain why a useful predictor transfers, but it is not an additional logical premise after coverage is given.

Why a null is not an impossibility result. A null benchmark row can arise because the true effects are small, the candidate is one-sided, the evidence map is weak, the benefit model underfits, the radius is wide, or calibration does not transfer. The population impossibility requires evidence-identical target laws inside the declared class with incompatible strict directions. The canonical benchmark table does not estimate that full fibre. It therefore reports weak or negative empirical regimes without calling them proven unknowable worlds.

What would falsify the practical claim. The practical claim is conditional and testable. A clean target replay with the locked pipeline can show excessive FA_u , poor coverage, or regret above both fixed policies. A checkpoint or adapter change can show that calibration does not transfer. A family-held-out split can show that cell-level cross-fitting was optimistic. Such failures do not refute the population identity; they refute the operational assumptions used to obtain a useful empirical interval.

K Formalization and Artifact Scope

The Lean project under `docs/research/kbound/formal/` provides a kernel-checked inventory for the theorem names it exports. The compact paper relies on the LaTeX statements above; it does not claim that every sentence has a one-to-one Lean theorem. In particular, deployment-class richness, risk alignment of a learned evidence map, exchangeability of a benchmark split, and correctness of data preprocessing remain outside the proof assistant unless separately encoded.

The current 65-name audit covers real algebra, finite uniform-rank implications, a two-atom target law, and conditional decision consequences. It does *not* derive exchangeable measurable scores' rank law, formalize a filtered nonnegative supermartingale and Ville maximal inequality, prove general KL/TV product-experiment Le Cam bounds, prove Hoeffding concentration, or derive `RichAt` for arbitrary deployment classes. The release's `-full-foundations` audit intentionally fails while these layers remain unmechanized.

The formal build and empirical build answer different questions. Lean checks that a proposition follows from formal assumptions. The reconciliation scripts check that displayed aggregates match saved records under implemented definitions. Neither checks the truth of an unencoded deployment assumption, and neither substitutes for independent experimental replication.

The authoritative compact artifacts are therefore layered:

1. this source and its compiled PDF for the scoped narrative;
2. shared theorem inputs for the exact statements and proofs;
3. canonical JSON and generated LaTeX for numerical results;
4. the source manifest for record-level provenance;
5. package tests and Lean logs for implementation and formal checks.

Historical drafts, dashboards, and exploratory win-search folders can remain useful development records, but they are not evidence sources for this compact submission unless referenced by the canonical manifest.

References

- [1] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, T. Darrell. Tent: Fully test-time adaptation by entropy minimization. *ICLR*, 2021.
- [2] J. Liang, D. Hu, J. Feng. Do we really need to access the source data? Source hypothesis transfer for unsupervised domain adaptation. *ICML*, 2020.
- [3] Y. Sun, X. Wang, Z. Liu, J. Miller, A. Efros, M. Hardt. Test-time training with self-supervision for generalization under distribution shifts. *ICML*, 2020.
- [4] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, M. Tan. Efficient test-time model adaptation without forgetting. *ICML*, 2022.
- [5] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, M. Tan. Towards stable test-time adaptation in dynamic wild world. *ICLR*, 2023.
- [6] Q. Wang, O. Fink, L. Van Gool, D. Dai. Continual test-time domain adaptation. *CVPR*, 2022.
- [7] M. Zhang, S. Levine, C. Finn. MEMO: Test time robustness via adaptation and augmentation. *NeurIPS*, 2022.
- [8] M. Schirmer, M. Jazbec, C. A. Naesseth, E. Nalisnick. Monitoring risks in test-time adaptation. *NeurIPS*, 2025. arXiv:2507.08721.
- [9] O. C. Mutlu, M. Honarmand, P. Azizian, S. Surabhi, D. P. Wall. Topology-guided test-time adaptation via persistent homology: From affective behavior analysis to autonomous driving. *CVPR Workshops*, pages 5331–5340, 2026.
- [10] A. H. Ariq. MMD-balls as credal sets: A PAC-Bayesian framework for epistemic uncertainty in test-time adaptation. arXiv:2605.21783, 2026.
- [11] A. Podkopaev, A. Ramdas. Tracking the risk of a deployed model and detecting harmful distribution shifts. *ICLR*, 2022. arXiv:2110.06177.
- [12] Y. Bar, S. Shaer, Y. Romano. Protected test-time adaptation via online entropy matching: A betting approach. *NeurIPS*, 2024. arXiv:2408.07511.
- [13] S. Garg, S. Balakrishnan, Z. Lipton, B. Neyshabur, H. Sedghi. Leveraging unlabeled data to predict out-of-distribution performance. *ICLR*, 2022. arXiv:2201.04234.
- [14] A. T. Kalai, V. Kanade. Towards optimally abstaining from prediction with OOD test examples. *NeurIPS*, 2021. arXiv:2105.14119.
- [15] J. Miller, R. Taori, A. Raghunathan, S. Sagawa, P. W. Koh, V. Shankar, P. Liang, Y. Carmon, L. Schmidt. Accuracy on the line: On the strong correlation between out-of-distribution and in-distribution generalization. *ICML*, 2021. arXiv:2107.04649.
- [16] C. Baek, Y. Jiang, A. Raghunathan, J. Z. Kolter. Agreement-on- the-line: Predicting the performance of neural networks under distribution shift. *NeurIPS*, 2022. arXiv:2206.13089.
- [17] T. Lee, S. Chottananurak, T. Gong, S.-J. Lee. AETTA: Label-free accuracy estimation for test-time adaptation. *CVPR*, 2024. arXiv:2404.01351.

- [18] J. Steinhardt, P. Liang. Unsupervised risk estimation using only conditional independence structure. *NeurIPS*, 2016. arXiv:1606.05313.
- [19] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, J. Wortman Vaughan. A theory of learning from different domains. *Machine Learning* 79(1–2):151–175, 2010.
- [20] E. Rosenfeld, S. Garg. (Almost) provable error bounds under distribution shift via disagreement discrepancy. *NeurIPS*, 2023. arXiv:2306.00312.
- [21] Y. Geifman, R. El-Yaniv. Selective classification for deep neural networks. *NeurIPS*, 2017.
- [22] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei. ImageNet: A large-scale hierarchical image database. *CVPR*, 2009.
- [23] D. Hendrycks, T. Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. *ICLR*, 2019.
- [24] P. W. Koh, S. Sagawa, H. Marklund, S. M. Xie, M. Zhang, A. Balsubramani, W. Hu, M. Yasunaga, et al. WILDS: A benchmark of in-the-wild distribution shifts. *ICML*, 2021.
- [25] P. Bándi, O. Geessink, Q. Manson, M. van Dijk, M. Balkenhol, M. Hermsen, et al. From detection of individual metastases to classification of lymph node status at the patient level: The CAMELYON17 challenge. *IEEE Trans. Med. Imaging* 38(2):550–560, 2019.
- [26] J. Taylor, S. L. Earnshaw, B. M. Mabey, M. Victors, M. N. Annis, D. Y. Torigian, et al. The RXXR1 dataset and challenge: Unsupervised classification of cellular morphological changes. *arXiv:1907.04758*, 2019.
- [27] H. Venkateswara, J. Eusebio, S. Chakraborty, S. Panchanathan. Deep hashing network for unsupervised domain adaptation. *CVPR*, 2017. (Office-Home dataset release).
- [28] I. Gulrajani, D. Lopez-Paz. In search of lost domain generalization. *ICLR*, 2021.
- [29] F. Croce, M. Andriushchenko, V. Schwag, E. Debenedetti, N. Flammarion, M. Chiang, P. Mittal, M. Hein. RobustBench: A standardized adversarial robustness benchmark. *NeurIPS Datasets and Benchmarks*, 2021.
- [30] B. Recht, R. Roelofs, L. Schmidt, V. Shankar. Do CIFAR-10 classifiers generalize to CIFAR-10? *ICML*, 2019.
- [31] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, et al. PyTorch: An imperative style, high-performance deep learning library. *NeurIPS*, 2019.
- [32] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al. Scikit-learn: Machine learning in Python. *JMLR* 12:2825–2830, 2011.
- [33] R. Xie, A. Odonnat, V. Feofanov, W. Deng, J. Zhang, B. An. MaNo: exploiting matrix norm for unsupervised accuracy estimation under distribution shifts. *NeurIPS*, 2024. arXiv:2405.18979.
- [34] W. Deng, Y. Suh, S. Gould, L. Zheng. Confidence and dispersity speak: characterizing prediction matrix for unsupervised accuracy estimation. *ICML*, 2023. arXiv:2302.01094.
- [35] J. Liang, R. He, T. Tan. A comprehensive survey on test-time adaptation under distribution shifts. *IJCV* 133(1):31–64, 2025. arXiv:2303.15361.
- [36] R. J. Tibshirani, R. F. Barber, E. Candès, A. Ramdas. Conformal prediction under covariate shift. *NeurIPS*, 2019. arXiv:1904.06019.

- [37] R. F. Barber, E. J. Candès, A. Ramdas, R. J. Tibshirani. Conformal prediction beyond exchangeability. *Annals of Statistics* 51(2):816–845, 2023. arXiv:2202.13415.
- [38] I. Gibbs, E. Candès. Adaptive conformal inference under distribution shift. *NeurIPS*, 2021. arXiv:2106.00170.
- [39] S. Park, E. Dobriban, I. Lee, O. Bastani. PAC prediction sets under covariate shift. *ICLR*, 2022. arXiv:2106.09848.
- [40] S. Bates, A. Angelopoulos, L. Lei, J. Malik, M. I. Jordan. Distribution-free, risk-controlling prediction sets. *JACM* 68(6), 2021. arXiv:2101.02703.
- [41] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei. Learn then test: calibrating predictive algorithms to achieve risk control. *Annals of Applied Statistics*, 2025. arXiv:2110.01052.
- [42] T. Hoang, D. M. Vo, M. N. Do. Persistent test-time adaptation in recurring testing scenarios. *NeurIPS*, 2024. arXiv:2311.18193.
- [43] J. Lee, D. Jung, S. Lee, J. Park, J. Shin, U. Hwang, S. Yoon. Entropy is not enough for test-time adaptation: from the perspective of disentangled factors. *ICLR*, 2024. arXiv:2403.07366.
- [44] T. Lim, J. Hwang, K. Lee. When and where to reset matters for long-term test-time adaptation. *ICLR*, 2026. arXiv:2603.03796.
- [45] S. Sonoda, Y. Mizuno, K. Tsukamoto, N. Onda. Lean formalization of generalization error bound by Rademacher complexity and Dudley’s entropy integral. 2025. arXiv:2503.19605.
- [46] S. Sreeram, Y. D. Kwon, C. Mascolo. Tempora: Characterising the time-contingent utility of online test-time adaptation. *ICML*, 2026. arXiv:2602.06136.
- [47] A. Corrada-Emmanuel. The logic of NTQR evaluations of noisy AI agents: Complete postulates and logically consistent error correlations. 2023. arXiv:2312.05392.
- [48] C. Gupta, A. Ramdas. Top-label calibration and multiclass-to-binary reductions. *ICLR*, 2022. arXiv:2107.08353.
- [49] S. Ben-David, R. Urner. On the hardness of domain adaptation and the utility of unlabeled target samples. *ALT*, 2012.
- [50] Z. C. Lipton, Y.-X. Wang, A. Smola. Detecting and correcting for label shift with black box predictors. *ICML*, 2018.
- [51] S. Garg, Y. Wu, S. Balakrishnan, Z. C. Lipton. A unified view of label shift estimation. *NeurIPS*, 2020.
- [52] C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Trans. Information Theory*, 16(1):41–46, 1970.
- [53] C. Cortes, G. DeSalvo, M. Mohri. Learning with rejection. *ALT*, 2016.
- [54] D. Madras, T. Pitassi, R. Zemel. Predict responsibly: improving fairness and accuracy by learning to defer. *NeurIPS*, 2018.
- [55] S. Beery, G. Van Horn, and P. Perona, “Recognition in Terra Incognita,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 456–473, doi: 10.1007/978-3-030-01270-0_28.